

Chapter 28

Organic Synthesis

&

Retrosynthetic Analysis

有机合成与逆合成分析

陆钊洪

参考书: Organic Chemistry, p695-722, Clayden, Greeves, Warren and Wothers, Oxford University Press, 2001

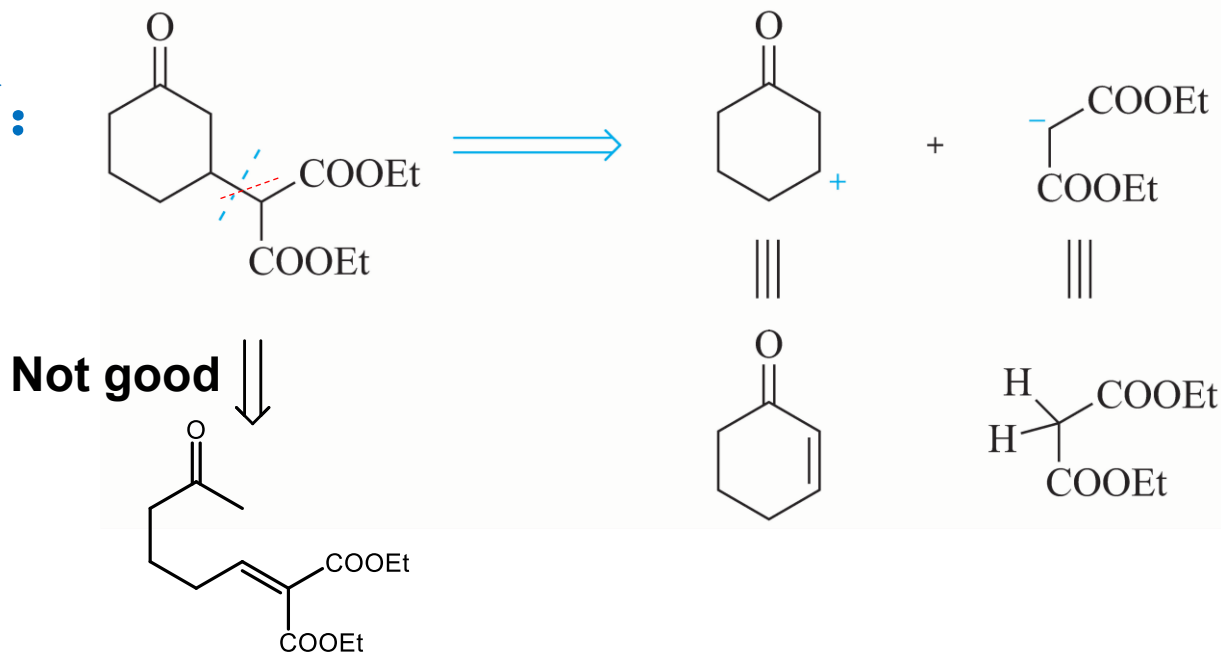




Retrosynthetic Analysis

20世纪60年代，E. J. Corey提出了逆合成分析理论。他从剖析目标化合物的分子结构入手，根据分子中各**原子间连接方式(化学键)的特征**，采用“**切断一根化学键**”分析法，综合运用有机化学反应方法和反应机理的知识，选择合适的化学键进行切断，**将目标分子逐渐转化成一些较小的中间体**；再以这些中间体作为新的目标分子，将其切断成更小的中间体；以此类推，直到找到可以方便购得的起始原料为止。

举例：





Criteria of Organic Synthesis

有机合成的基本要求包括：合成的反应步骤越少 (step economy) 越好，每步反应的产率越高 (efficient) 越好，以及原料越便宜越易得越好。原子利用率(atom economy)高、反应条件温和以及绿色化学(environment-benign, 环境友好)是现代有机合成的最高目标。

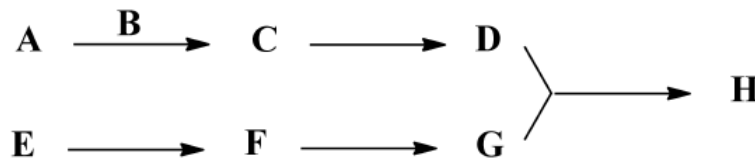
合成路线设计：线性合成路线 V.S. 汇聚式合成路线

- 线性 (linear) 合成路线：



如果每步产率为90%，6步反应后总产率为53%。

- 汇聚式 (convergent) 合成路线：



如果每步产率为90%，两步的总产率为81%；再汇集成合成后的产率为73%。与线性合成相比，汇聚式合成不仅反应步骤要少一步，而且产率提高了很多。

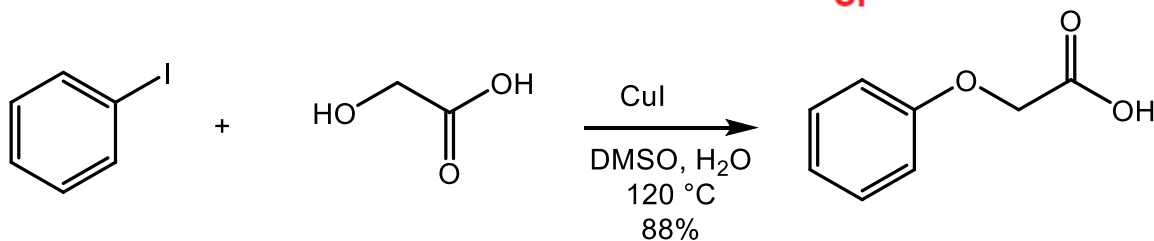
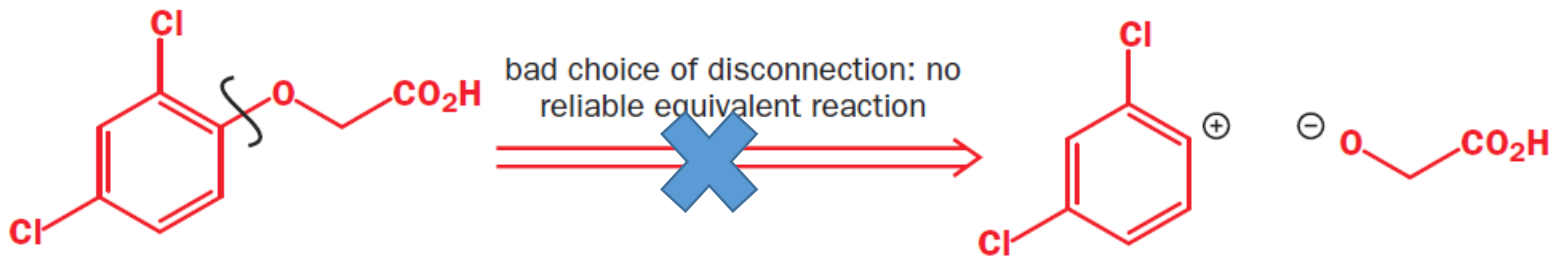
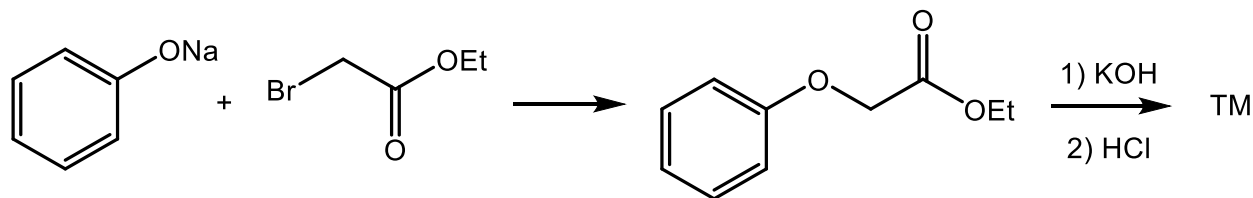
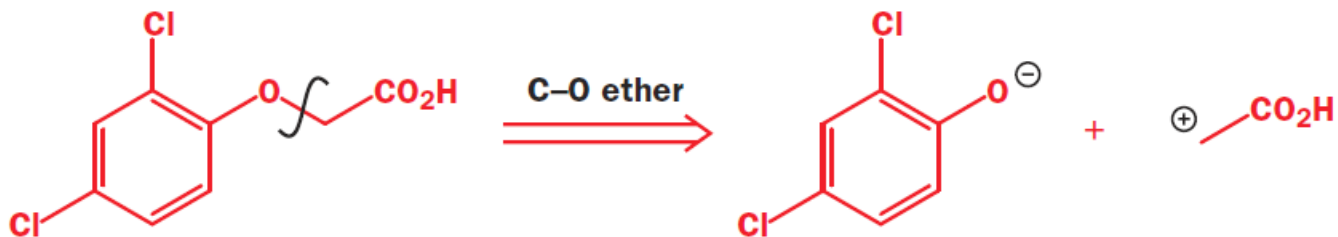


合理的合成设计的基本要素：
正确的化学反应性
区域选择性
立体选择性



Choosing Disconnection

- Guideline 1: Disconnections must correspond to known, reliable reactions (选择可靠的反应)



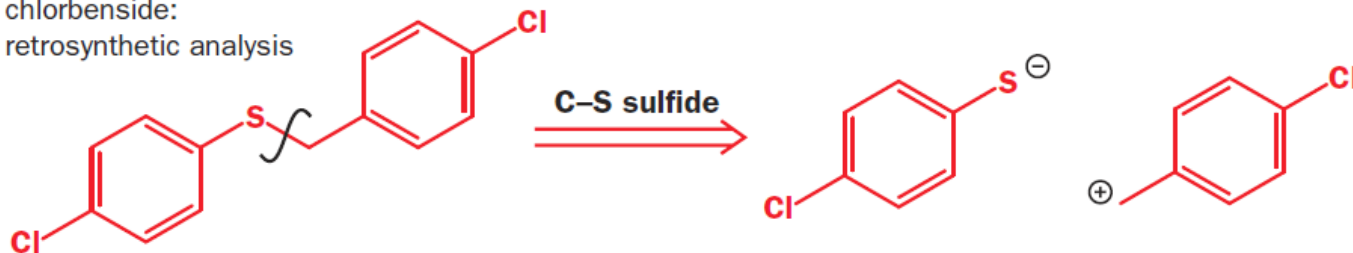


Choosing Disconnection

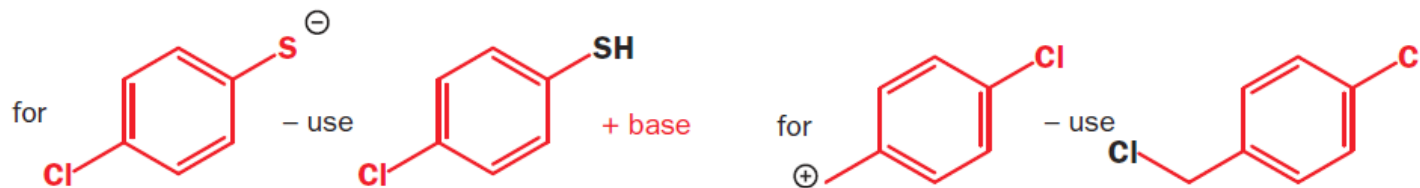
- Guideline 2: For compounds consisting of two parts joined by a **heteroatom**, disconnect next to the heteroatom

杂原子处断裂法

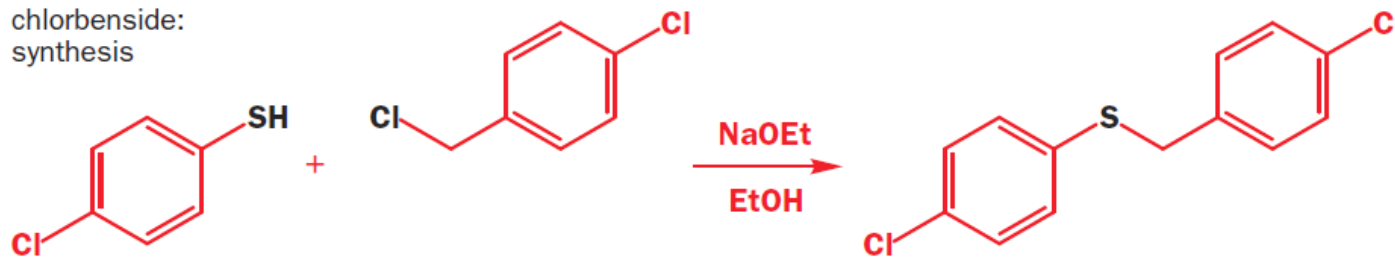
chlorbendside:
retrosynthetic analysis



We can now suggest reagents corresponding to the synthons, and propose a synthetic scheme.



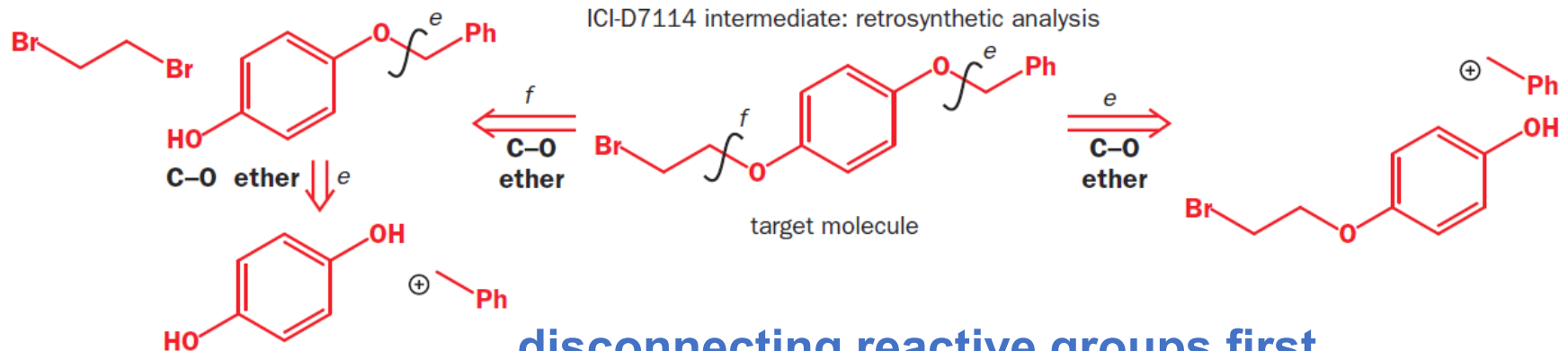
chlorbendside:
synthesis





- ICI-D7114 intermediate: retrosynthetic analysis

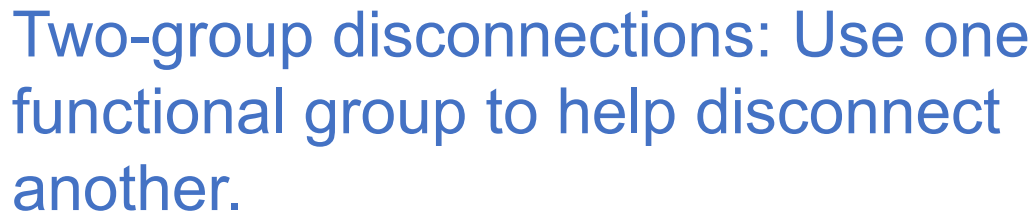
ICI-D7114 intermediate: retrosynthetic analysis



disconnecting reactive groups first



- ## 化学选择 性问题



is:

Reaction scheme showing the synthesis of 4-phenyl-2-methylpent-3-en-2-ol:

Starting material: 3-methylbut-3-en-1-ol

Step 1: Reaction with NaH to form the alkoxide intermediate.

Step 2: Reaction with 2-phenyloxirane (propylene oxide) to form the final product, 4-phenyl-2-methylpent-3-en-2-ol.

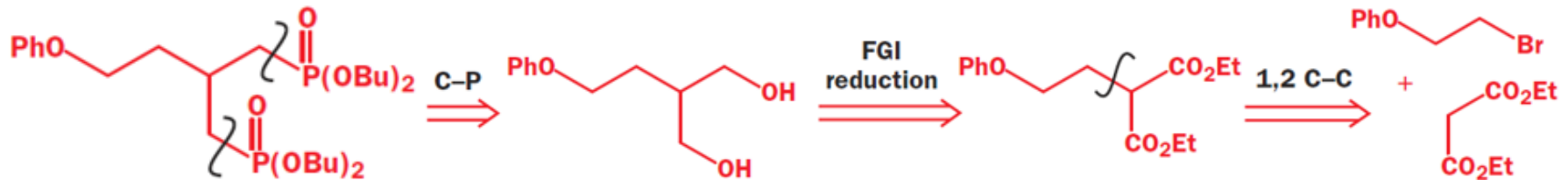
其他断法？



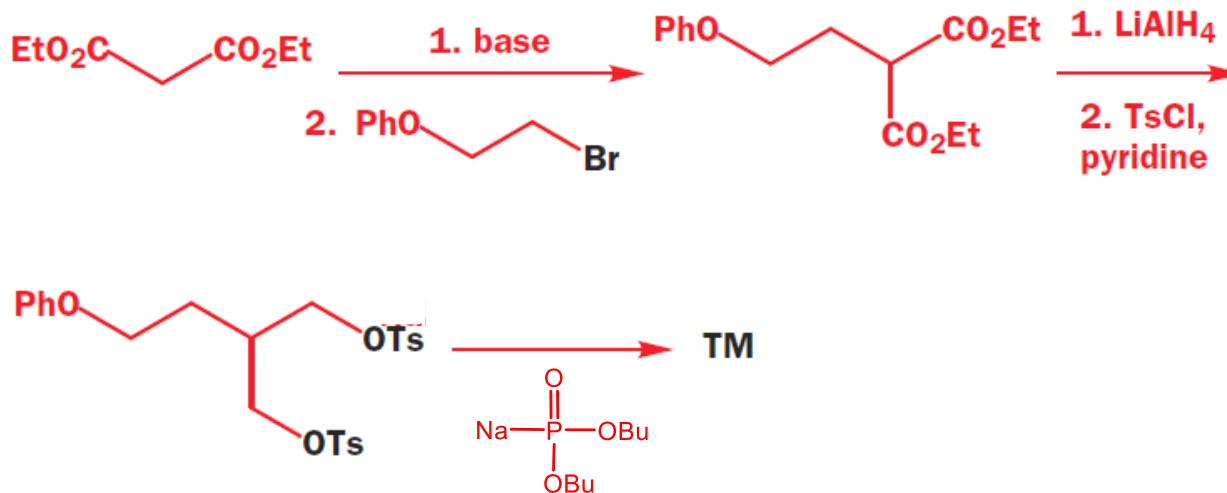
Choosing Disconnection

- Guideline 5: Convert to **oxygen-based functional groups** to facilitate C–C disconnections

belfosil: retrosynthetic analysis



belfosil: synthesis



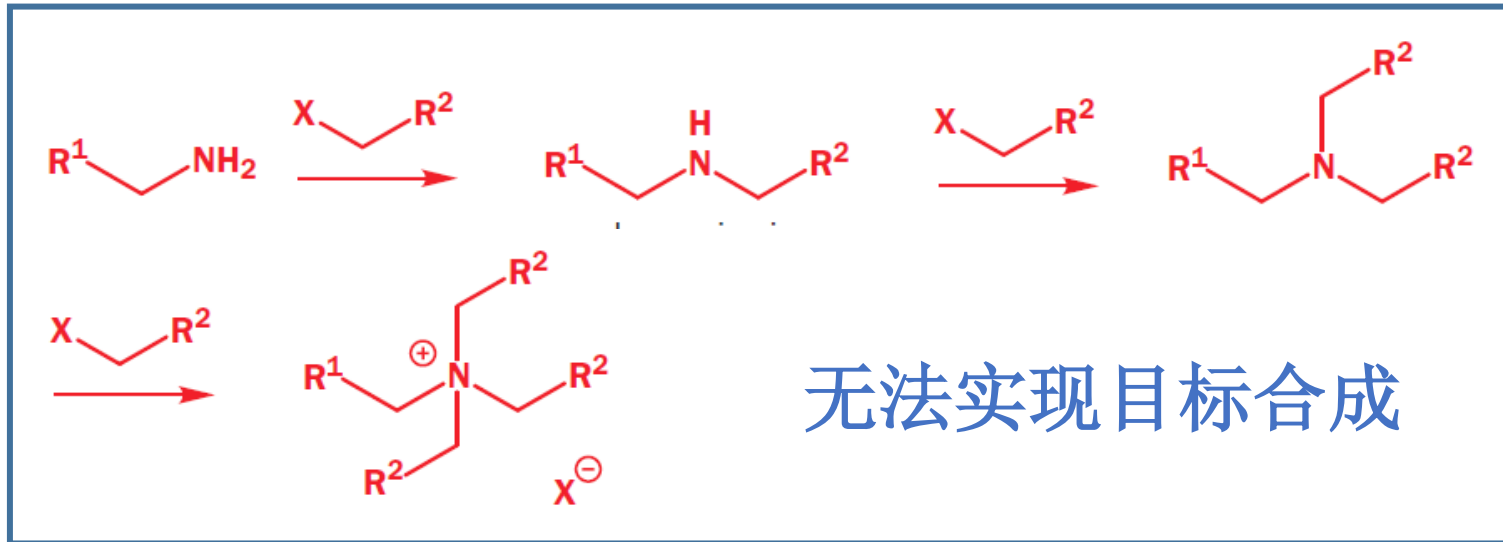
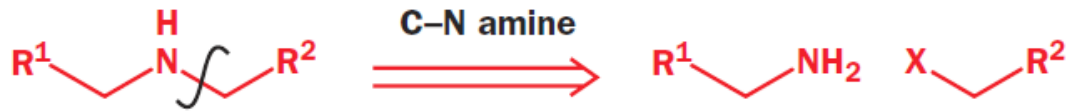


Selected Examples for Different Synthetic Strategies

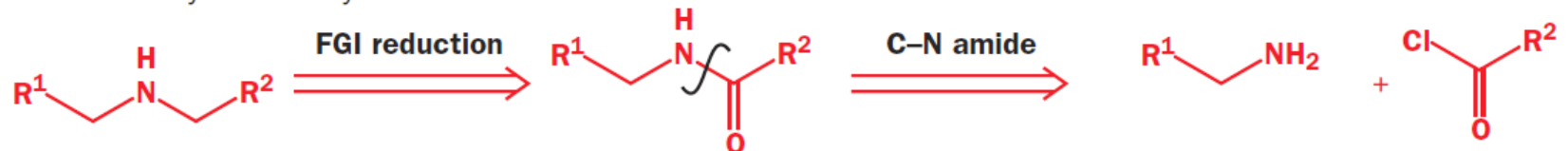
常见合成策略举例



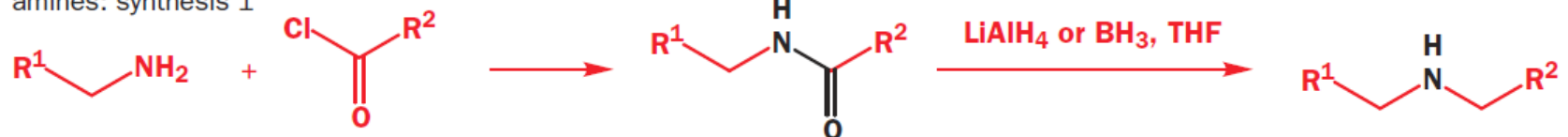
Functional Group Interconversion



amines: retrosynthetic analysis 1



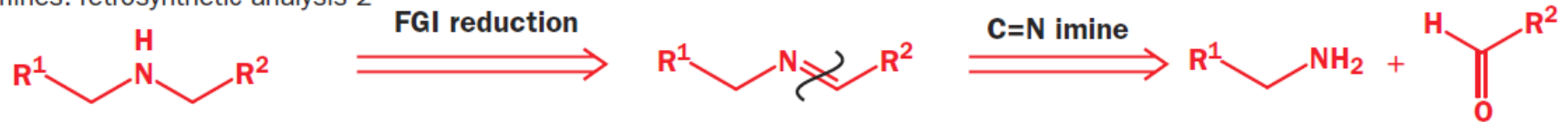
amines: synthesis 1



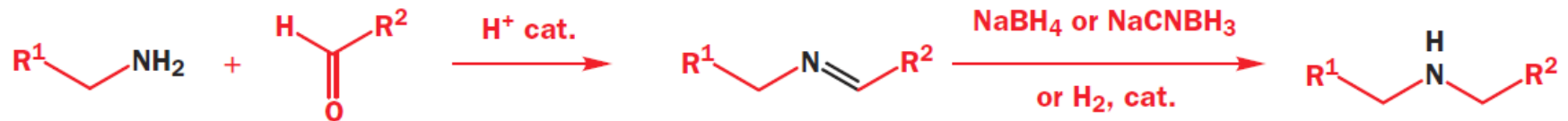


Functional Group Interconversion

amines: retrosynthetic analysis 2

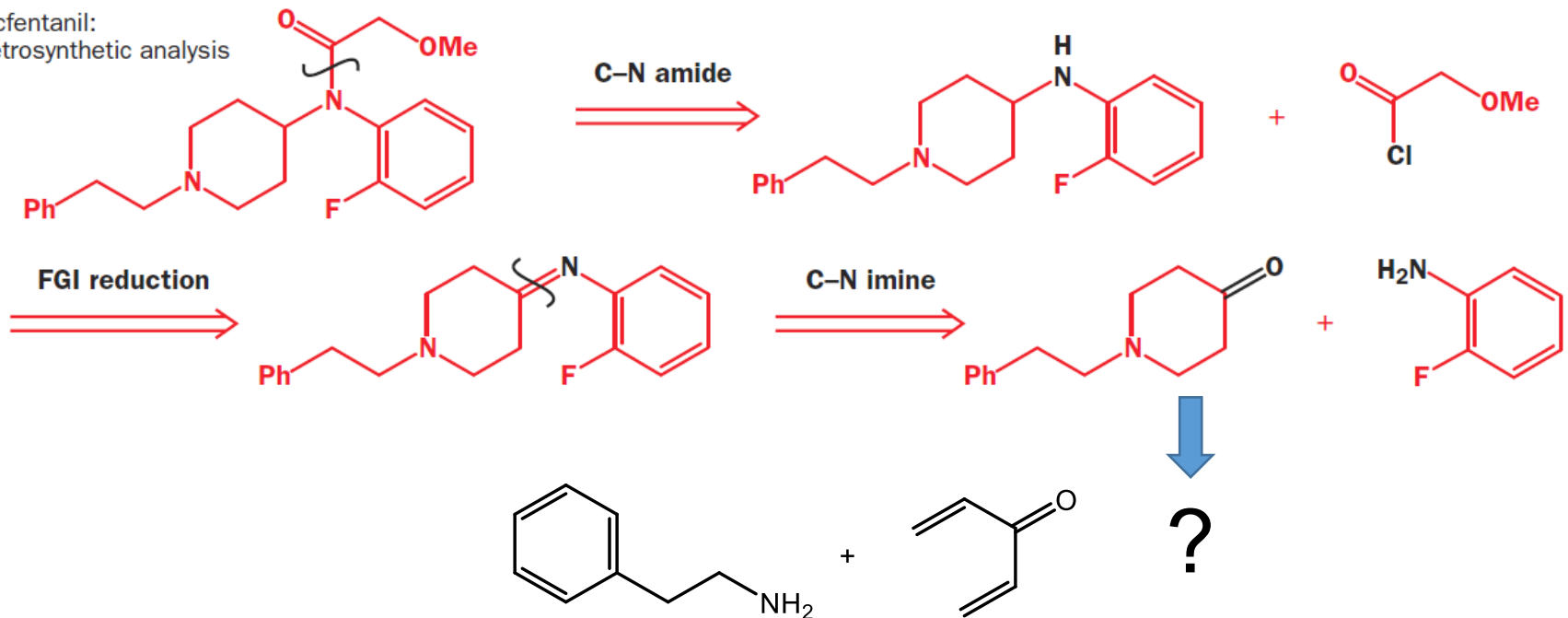


amines: synthesis 2 (reductive amination)



Example:

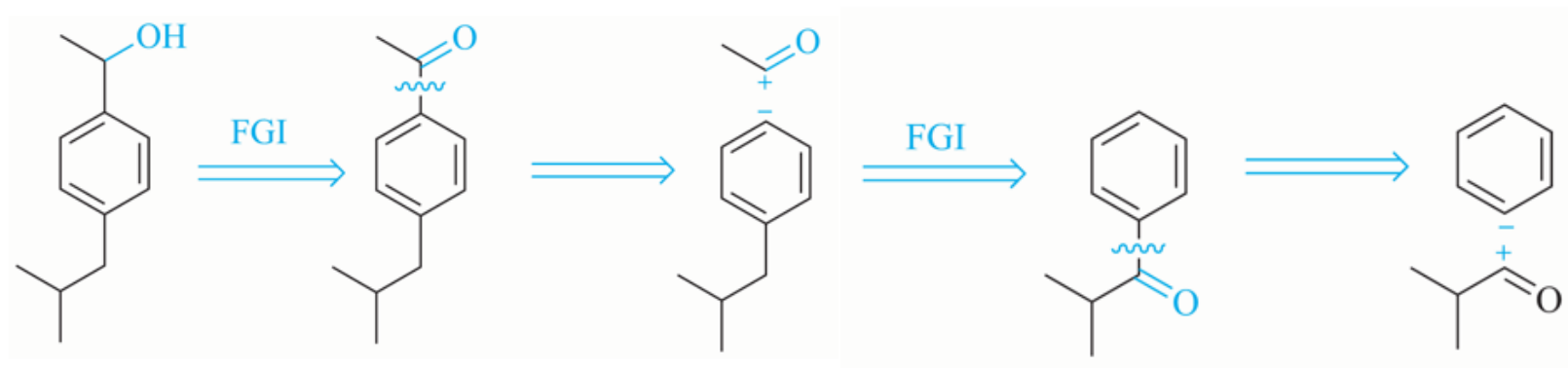
ocfentanil:
retrosynthetic analysis



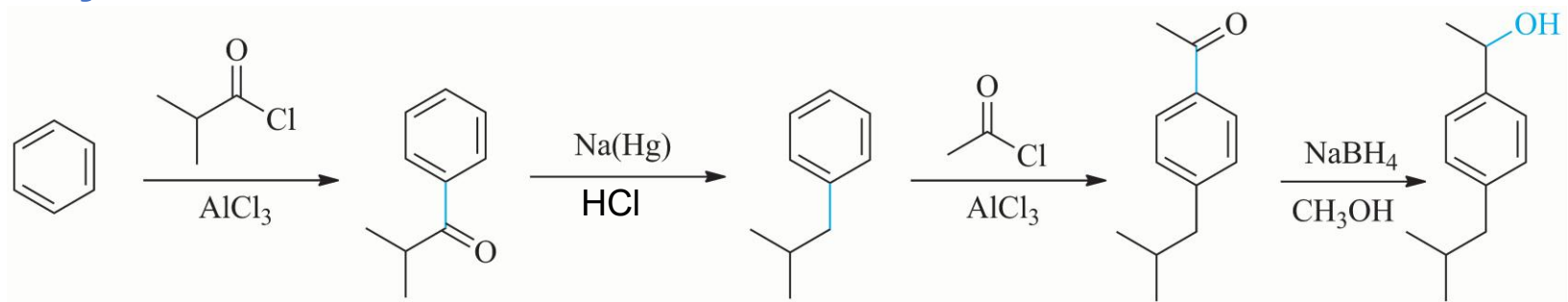


Functional Group Interconversion

Retrosynthetic analysis:



Synthesis:

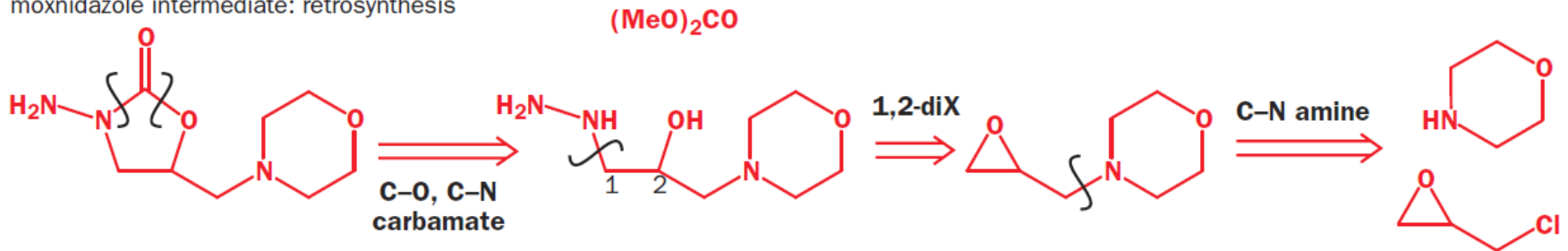




Two-group disconnections

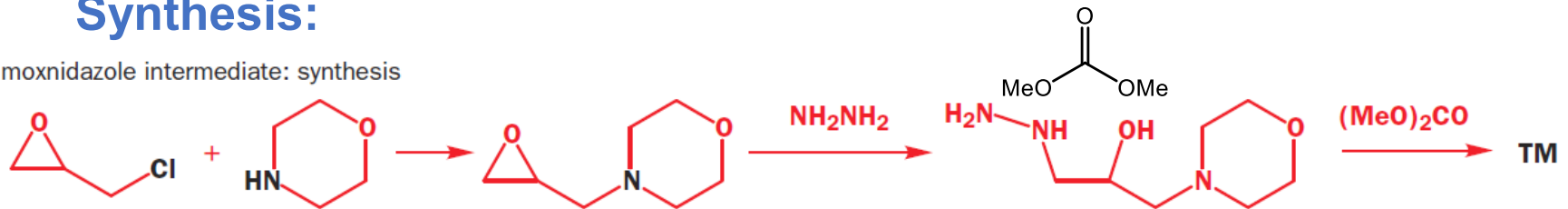
Retrosynthetic analysis:

moxnidazole intermediate: retrosynthesis



Synthesis:

moxnidazole intermediate: synthesis

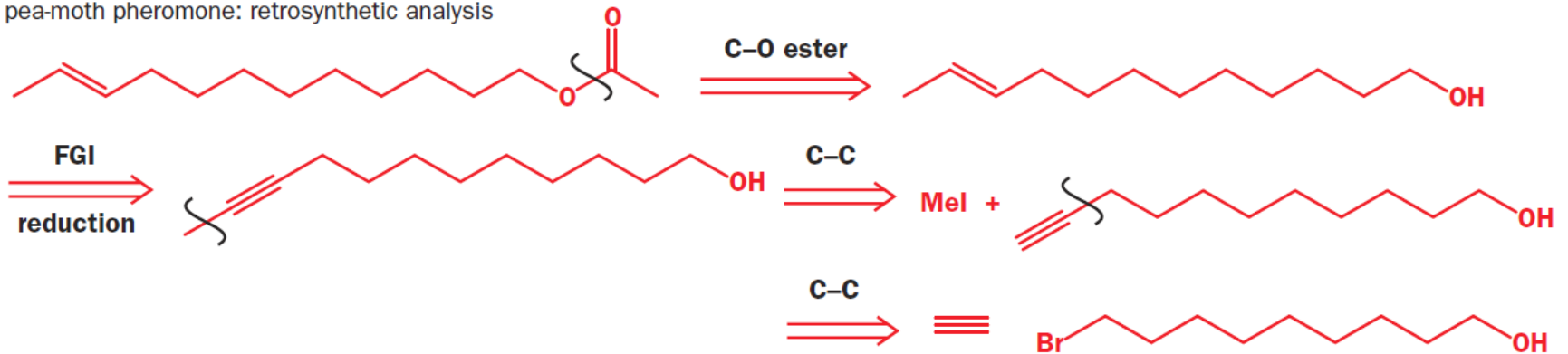




C–C disconnections

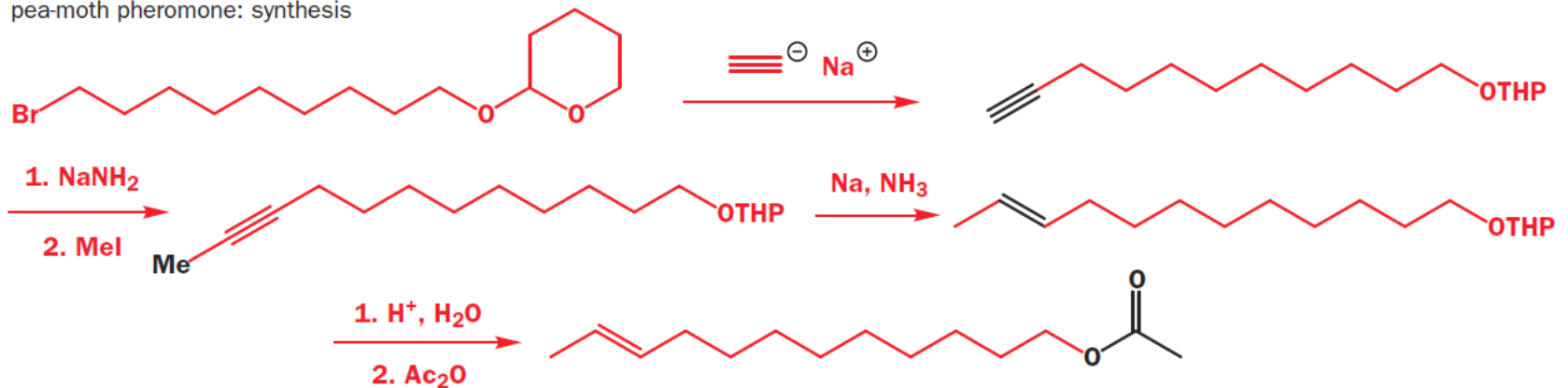
Retrosynthetic analysis:

pea-moth pheromone: retrosynthetic analysis



Synthesis:

pea-moth pheromone: synthesis

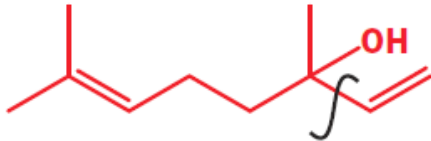




C–C disconnections

Retrosynthetic analysis:

linalool: retrosynthetic analysis

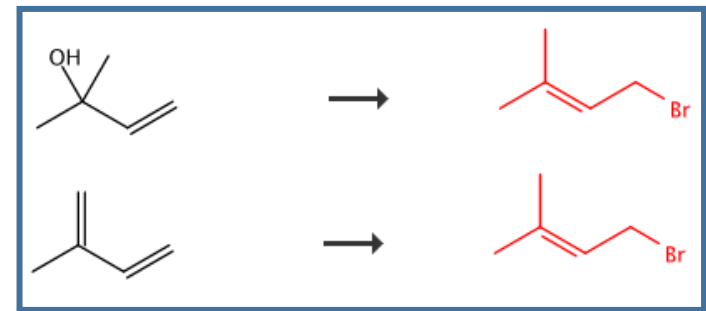
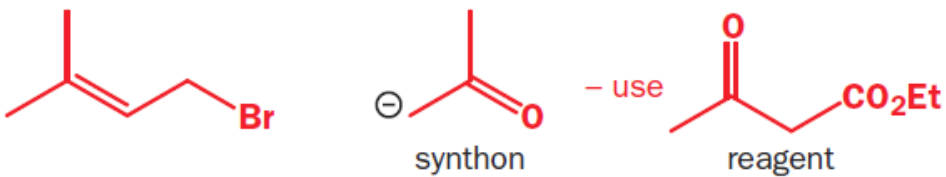
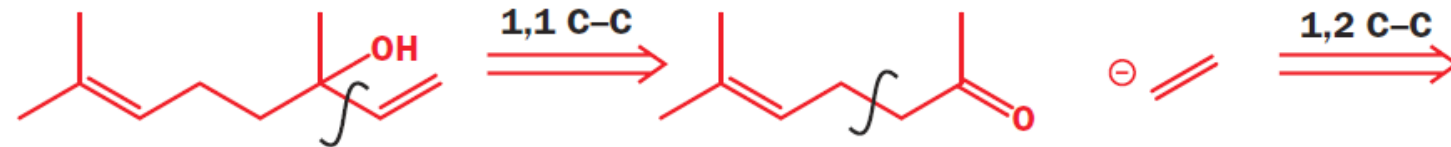




C-C disconnections

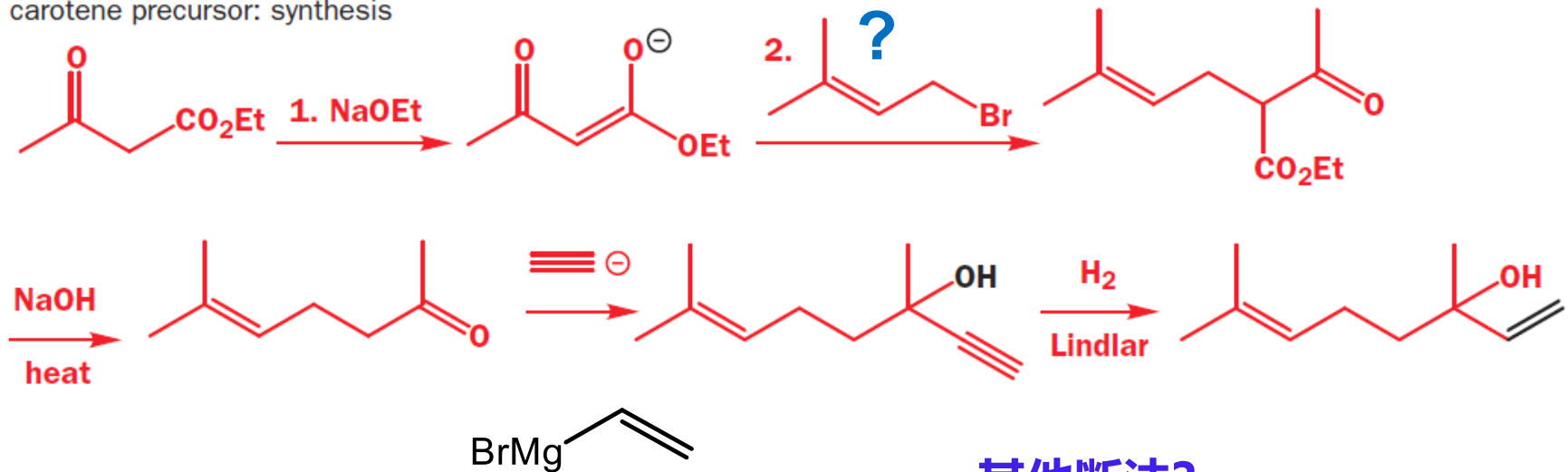
Retrosynthetic analysis:

linalool: retrosynthetic analysis



Synthesis:

carotene precursor: synthesis



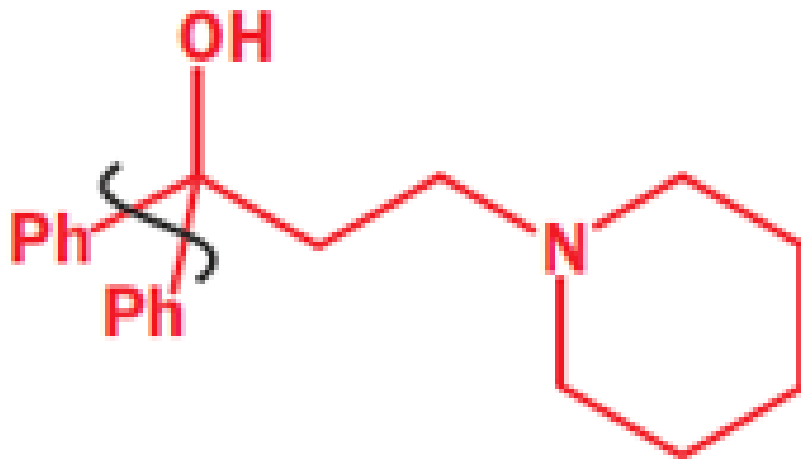
其他断法?



C–C disconnections

Retrosynthetic analysis:

fenpiprane precursor: retrosyn

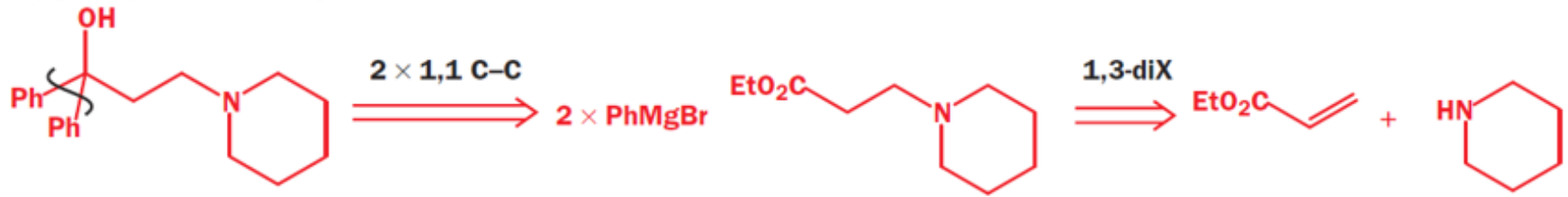




C–C disconnections

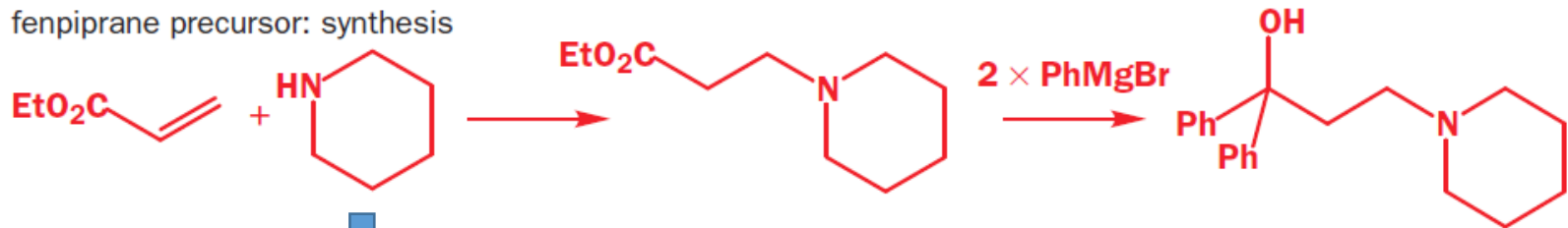
Retrosynthetic analysis:

fenpiprane precursor: retrosyn



Synthesis:

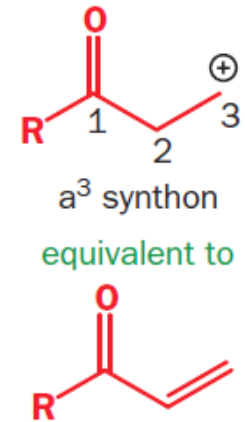
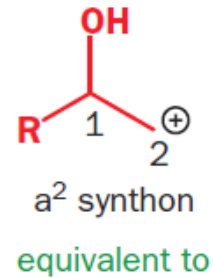
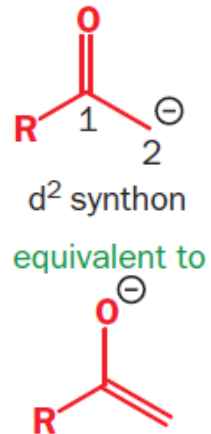
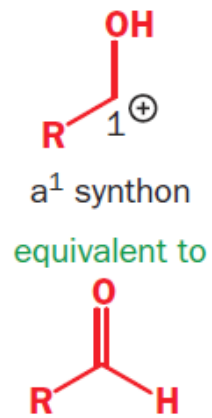
fenpiprane precursor: synthesis



怎么合成?



Donor and acceptor synthons



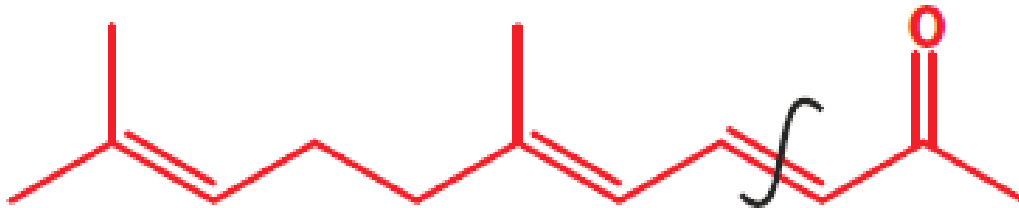
合成子等价体



C–C disconnections

Retrosynthetic analysis:

carotene intermediate: retrosynthetic analysis:





carotene intermediate: retrosynthetic analysis



carotene intermediate: synthesis

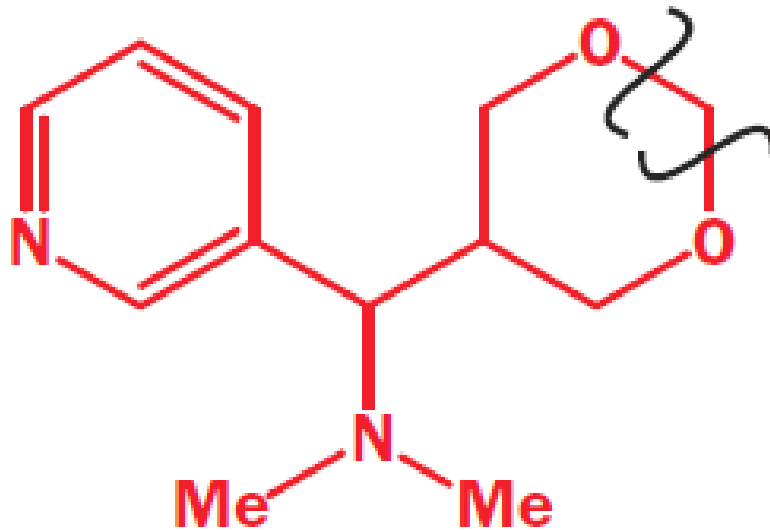




C–C disconnections

Retrosynthetic analysis:

doxpicomine: retrosynthetic ar

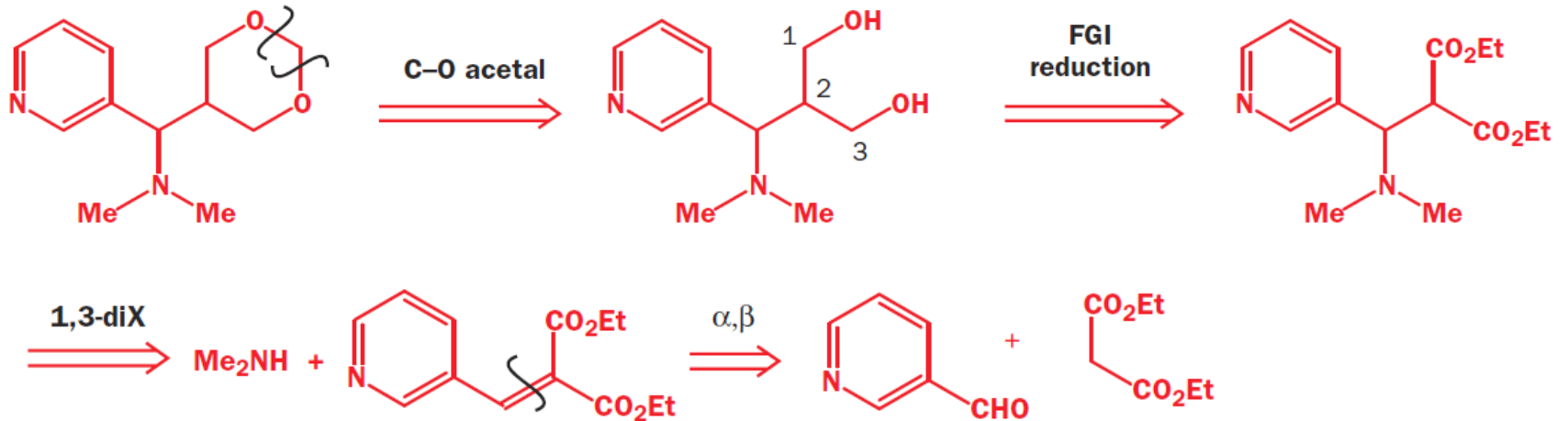




C–C disconnections

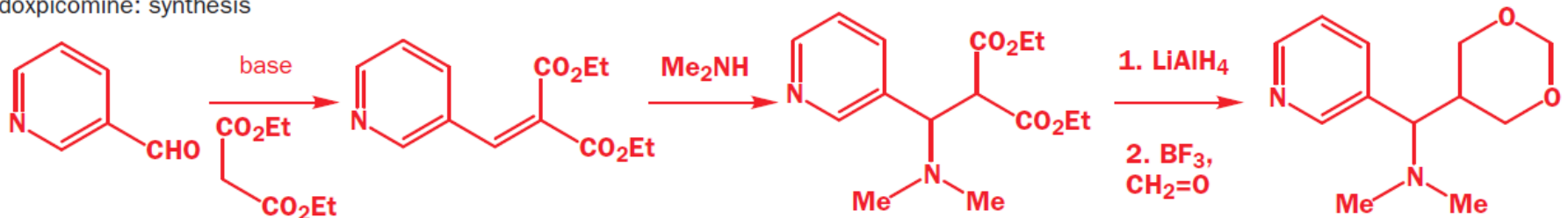
Retrosynthetic analysis:

doxpicomine: retrosynthetic analysis I



Synthesis:

doxpicomine: synthesis



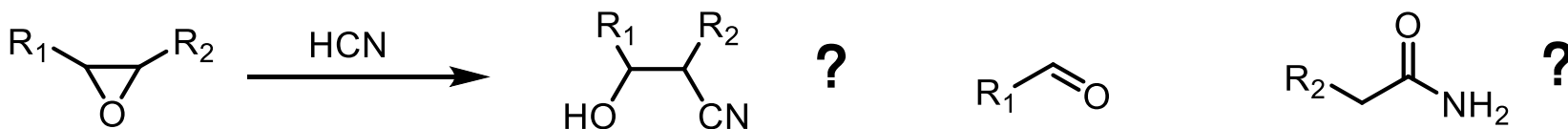
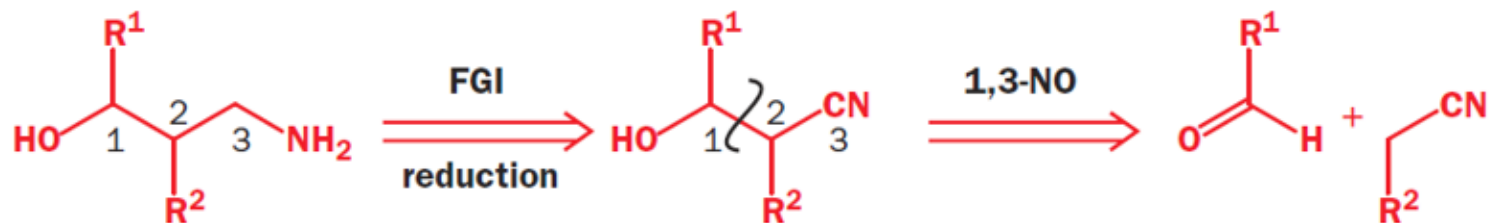
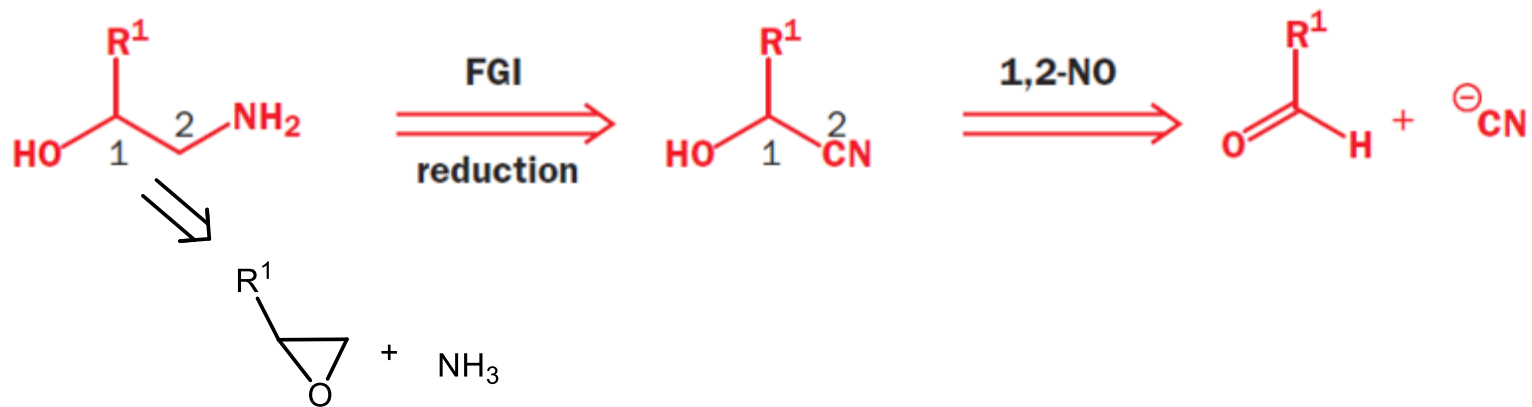
还有什么方案?



Aldol-style disconnections

Retrosynthetic analysis:

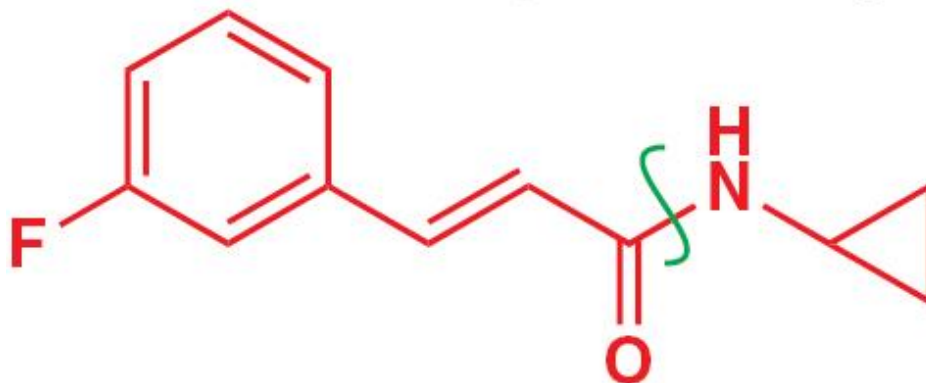
Synthesis of 1,2- or 1,3-amino alcohol





Aldol-style disconnections

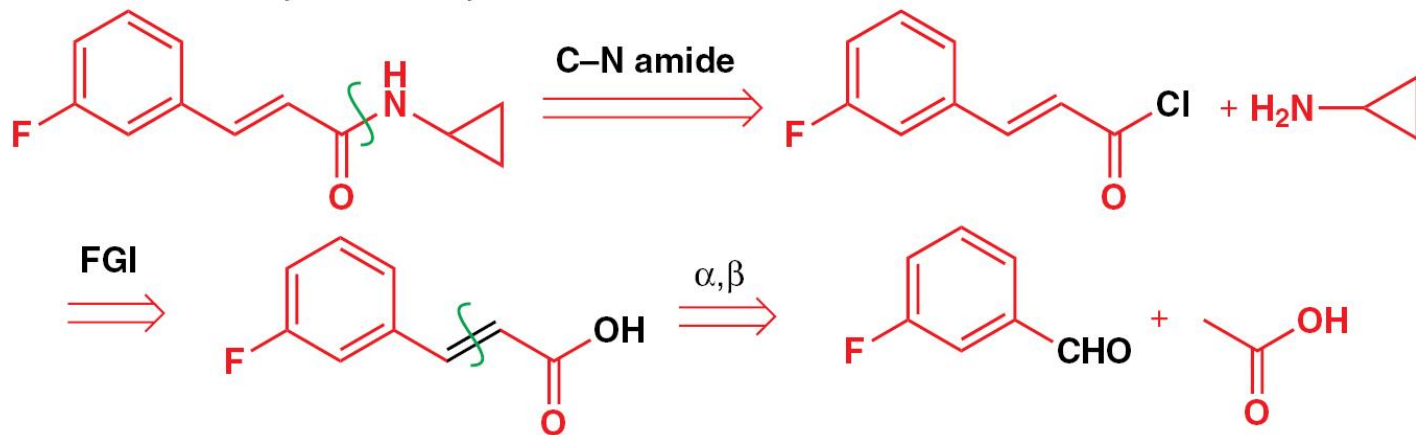
cinflumide: retrosynthetic analysis





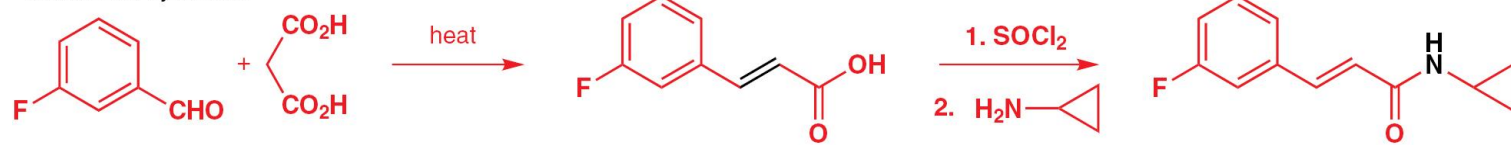
Aldol-style disconnections

cinflumide: retrosynthetic analysis





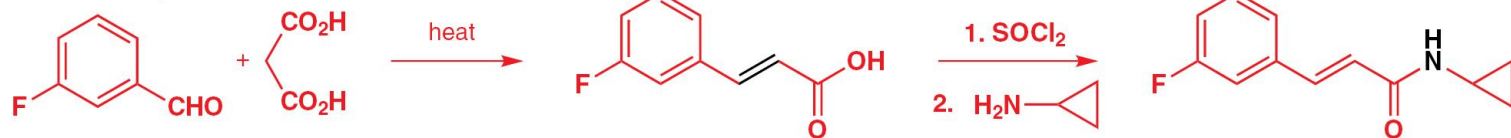
cinflumide: synthesis





Aldol-style disconnections

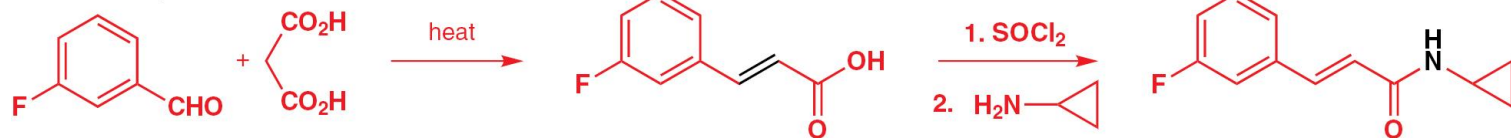
cinflumide: synthesis





Aldol-style disconnections

cinflumide: synthesis



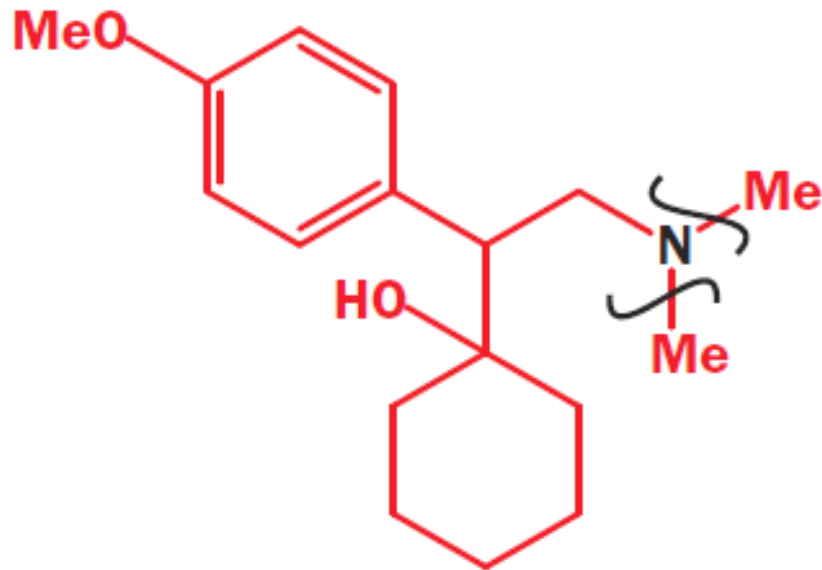


Aldol-style disconnections

Retrosynthetic analysis:

venlafaxine: retrosynthetic analysis

1,3-amino alcohol





venlafaxine: retrosynthetic analysis



venlafaxine: synthesis



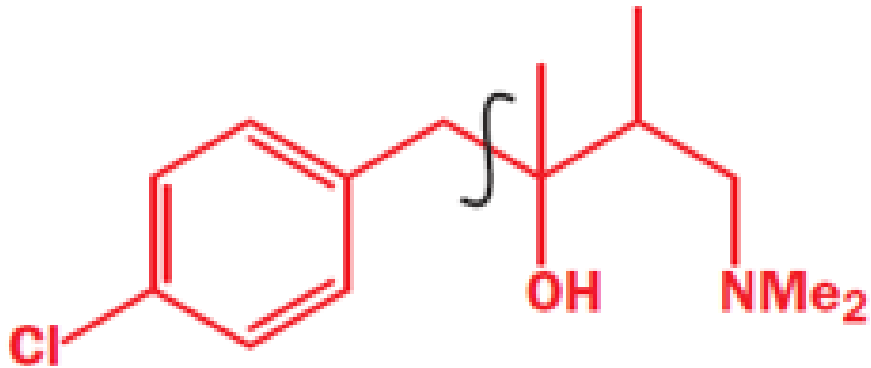


Aldol-style disconnections

Retrosynthetic analysis:

1,3-amino alcohol

clobutinol: retrosynthetic analysis

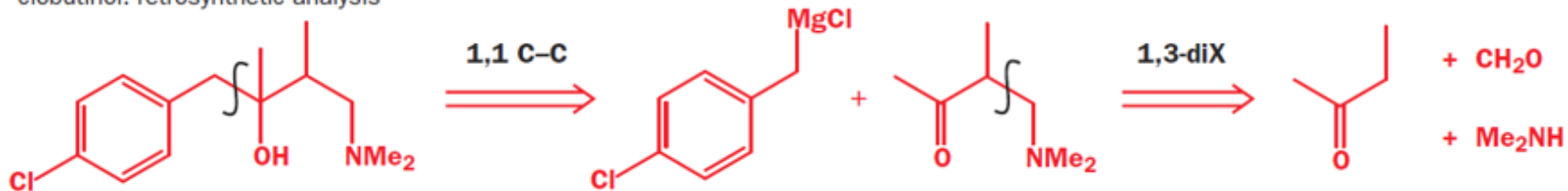




Aldol-style disconnections

Retrosynthetic analysis:

clobutinol: retrosynthetic analysis

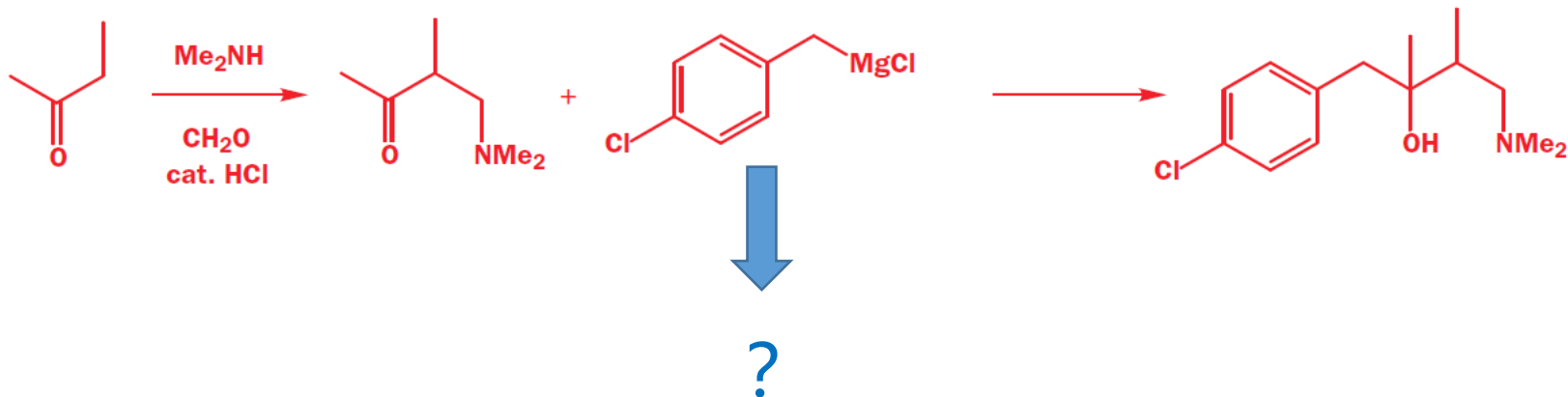


1,3-amino alcohol

Mannich disconnection

Synthesis:

clobutinol: synthesis

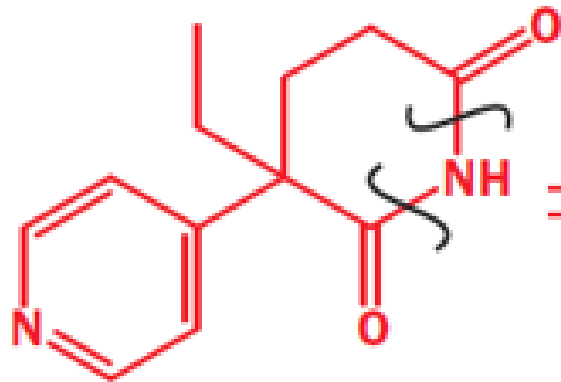




Aldol-style disconnections

Retrosynthetic analysis:

rogletimide: retrosynthetic analysis

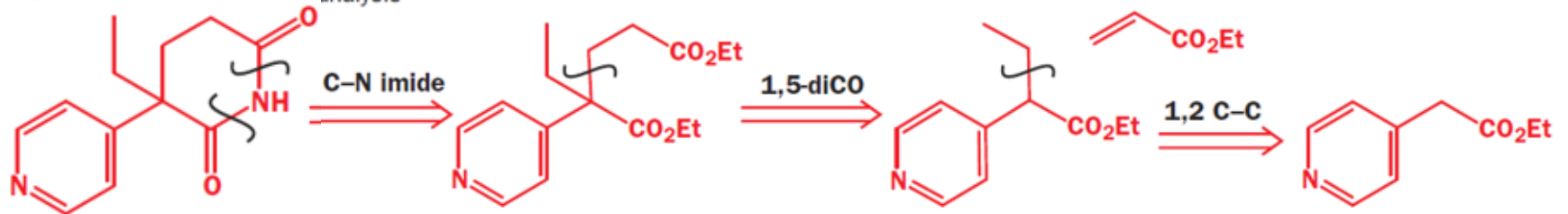




Aldol-style disconnections

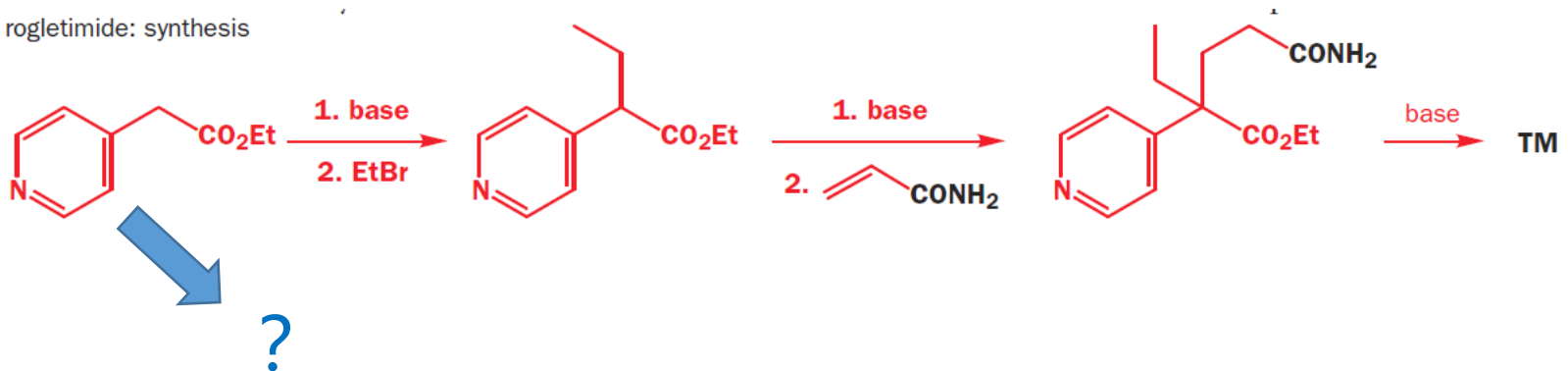
Retrosynthetic analysis:

rogletimide: retrosynthetic



Synthesis:

rogletimide: synthesis

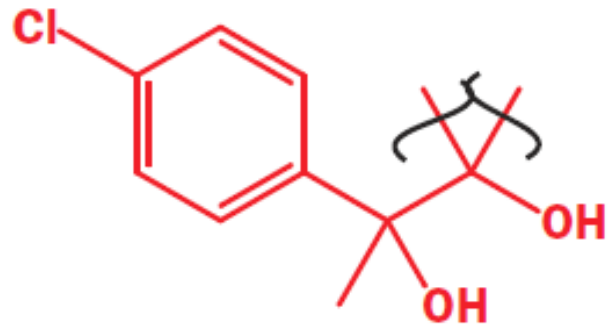




Umpolung 极性翻转

Retrosynthetic analysis:

phenaglycodol: retrosynthetic a

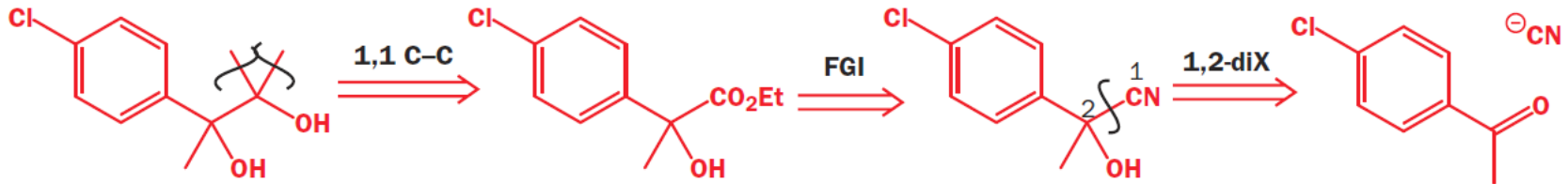




Umpolung 极性翻转

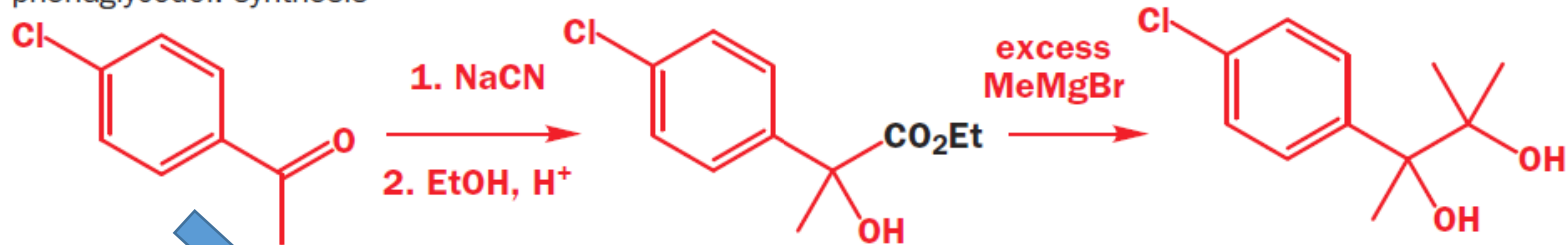
Retrosynthetic analysis:

phenaglycodol: retrosynthetic analysis



Synthesis:

phenaglycodol: synthesis



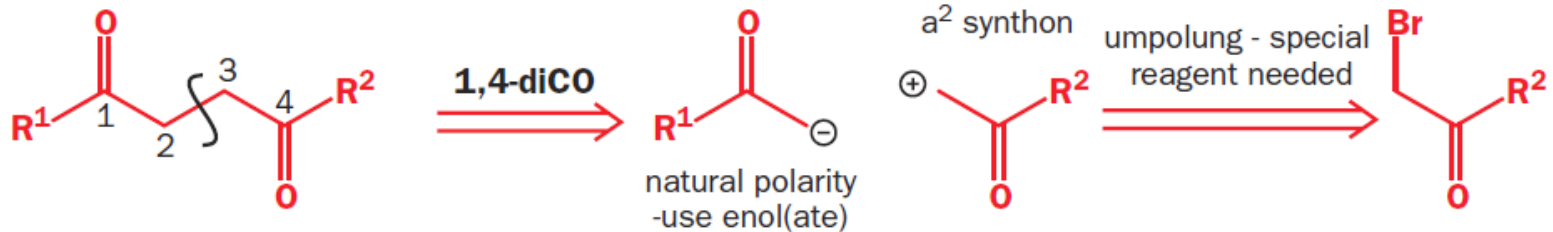
Pinacol coupling?

?

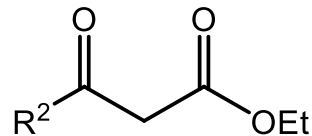


Umpolung 极性翻转

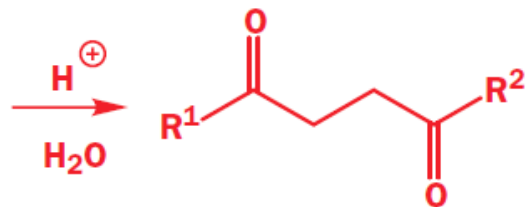
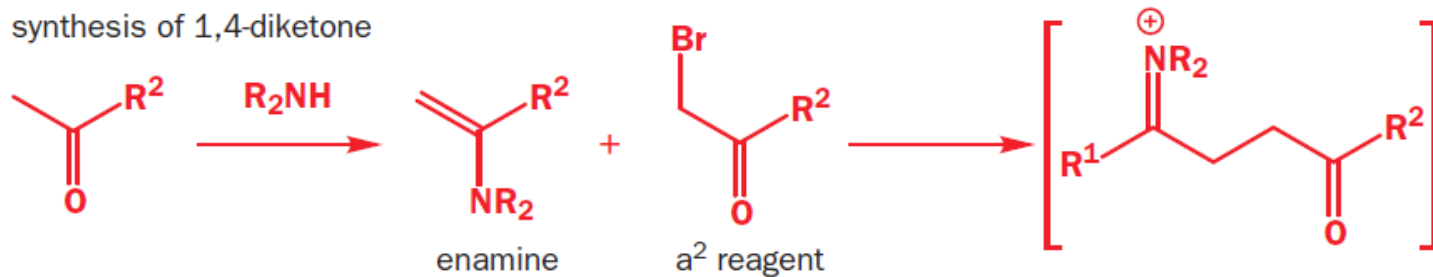
Retrosynthetic analysis:



Synthesis:



synthesis of 1,4-diketone



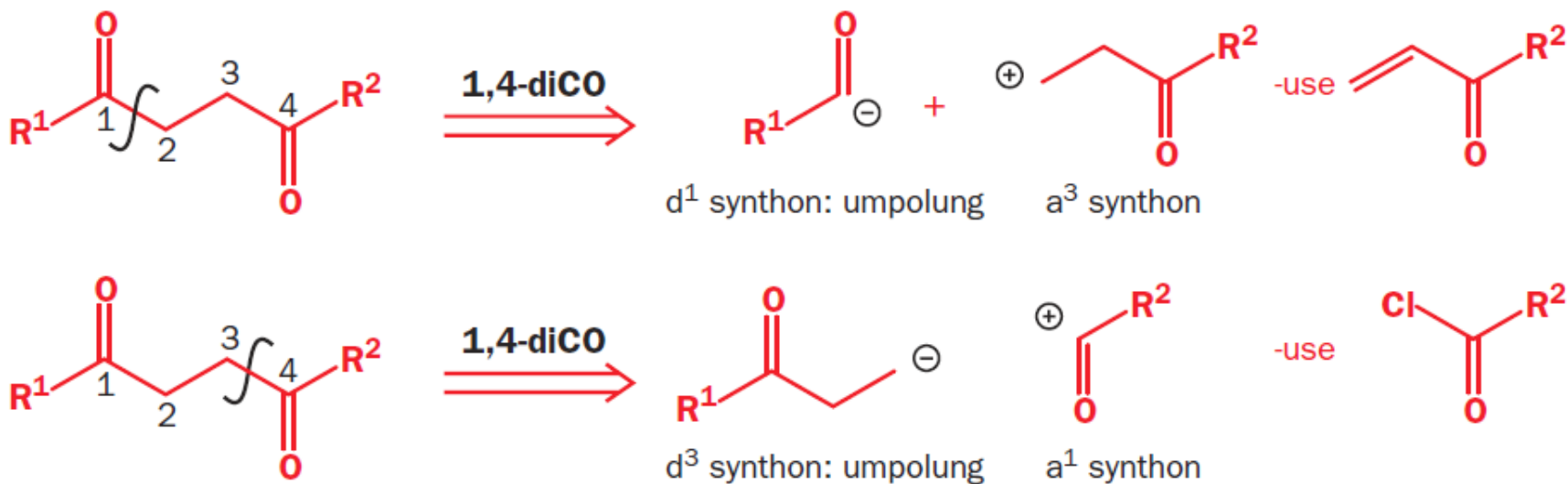
Stork烯胺化选择位阻小的一侧，避免空间排斥

甲基酮的制备方法？



Umpolung 极性翻转

Retrosynthetic analysis:

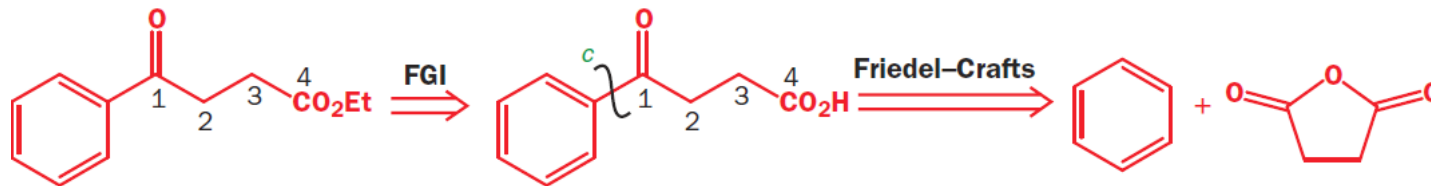
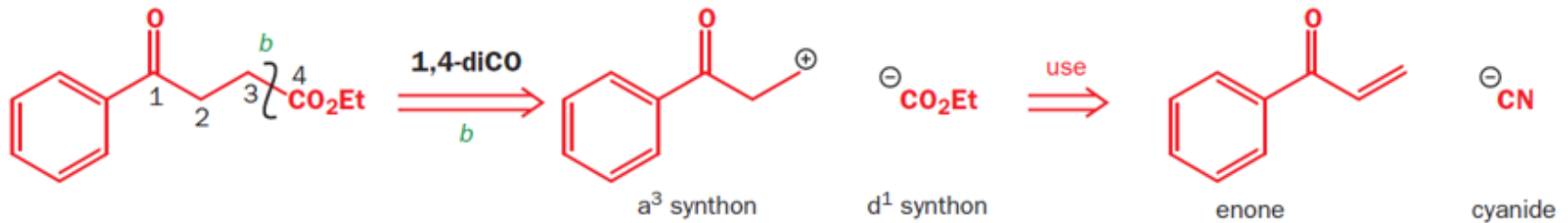
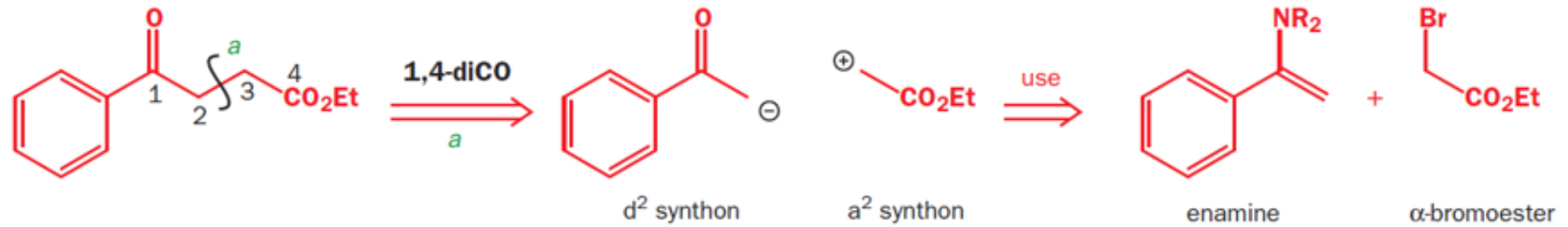


1,4-二羰基化合物的逆合成分析



Umpolung 极性翻转

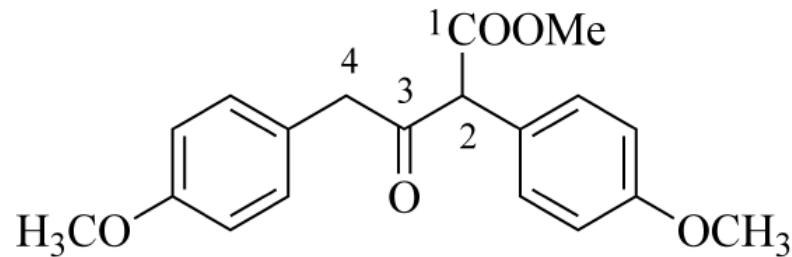
Examples:



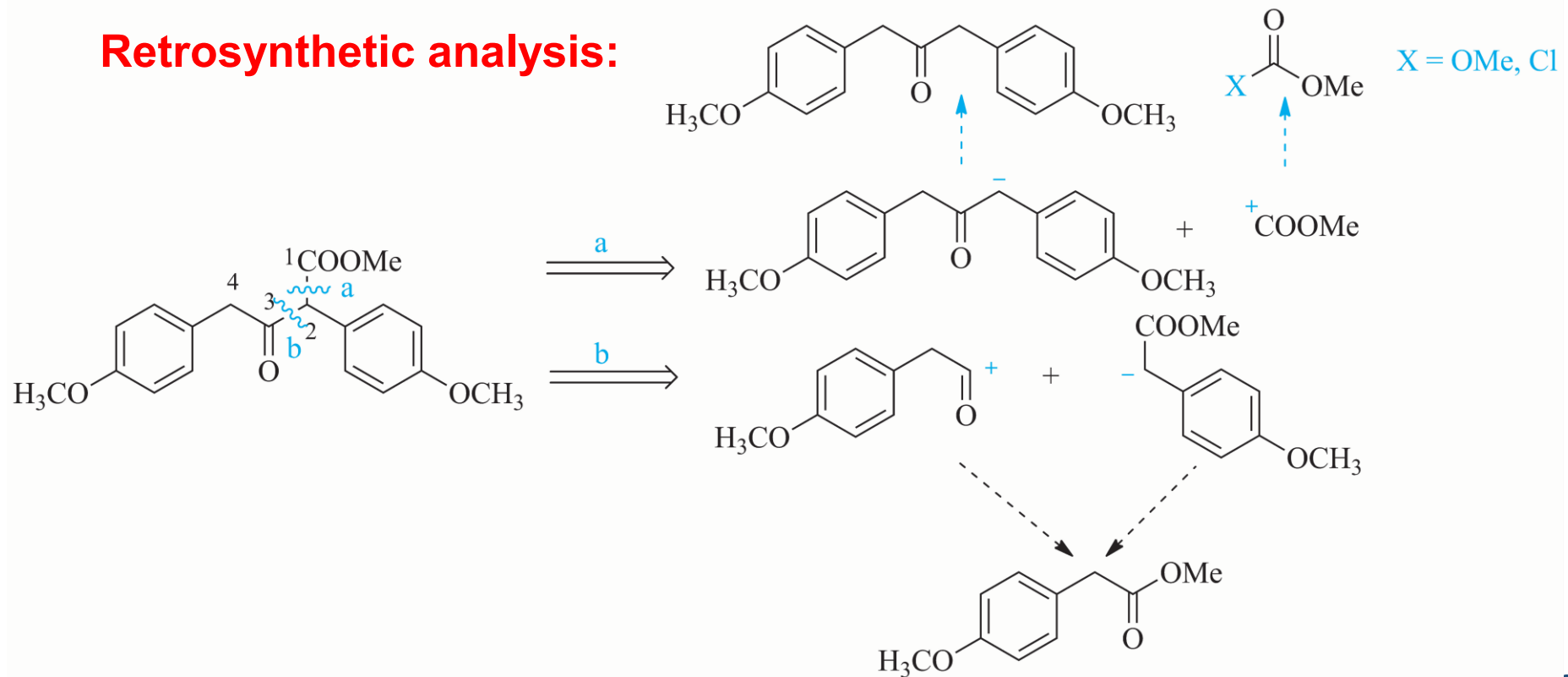


综合案例

一、以苯酚、甲醛和必要的无机试剂为原料合成：

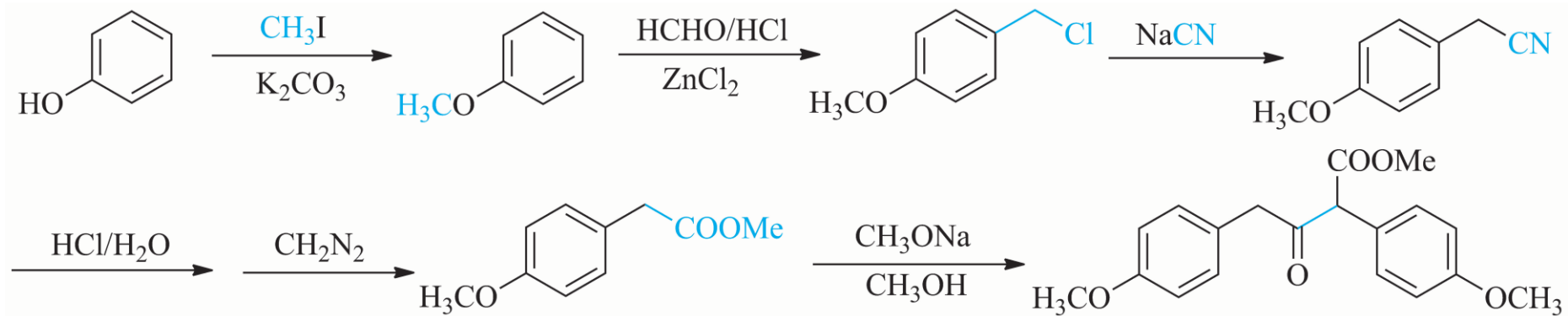


Retrosynthetic analysis:



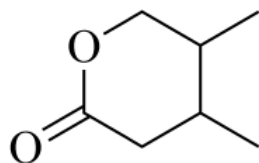


Synthesis:

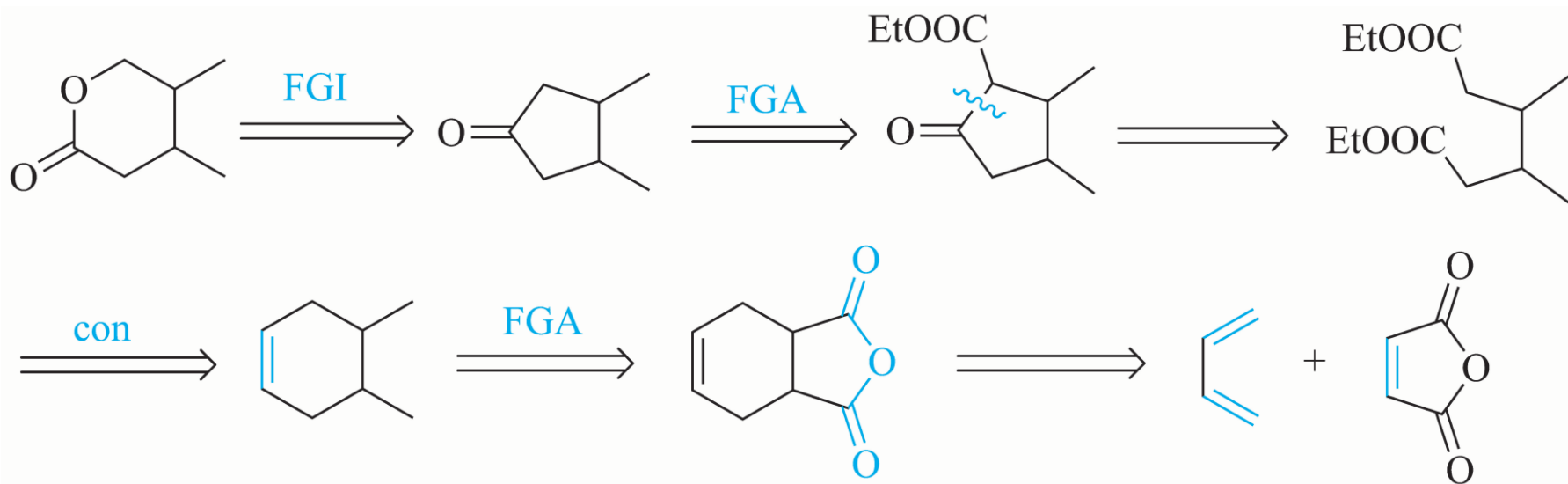




二、以1,3-丁二烯和顺丁烯二酸酐为原料合成:



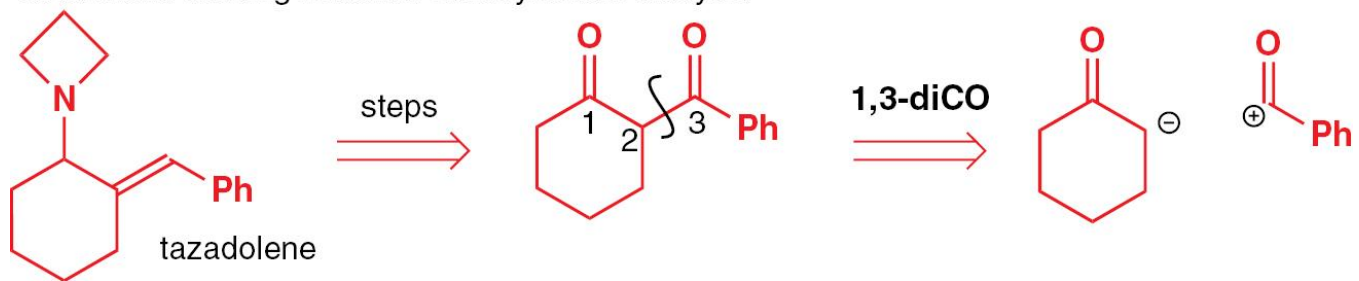
Retrosynthetic analysis:

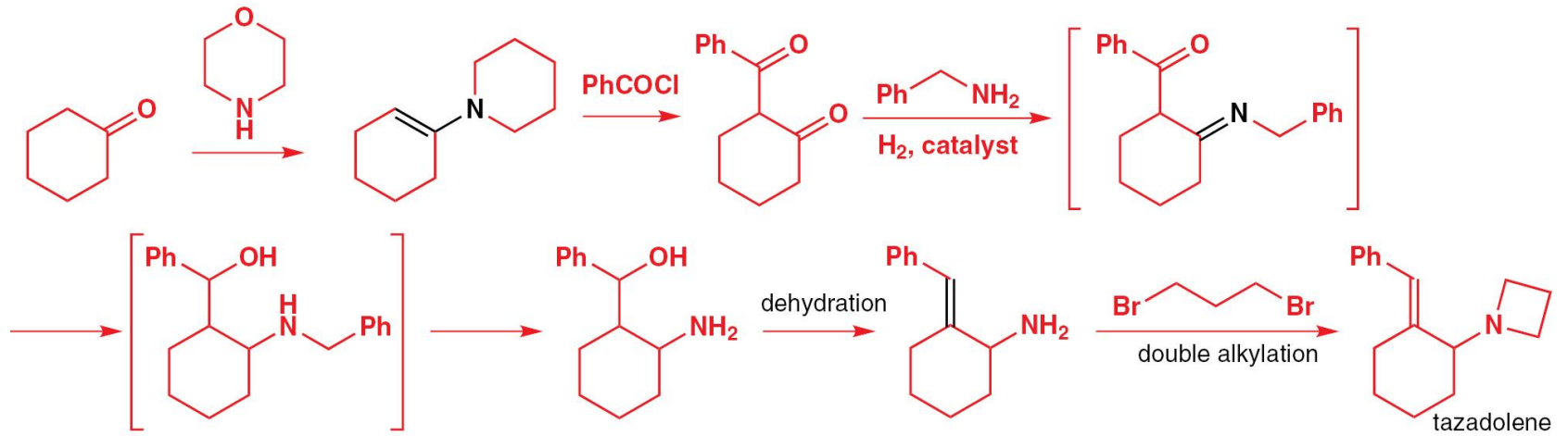






tazadolene starting material: retrosynthetic analysis





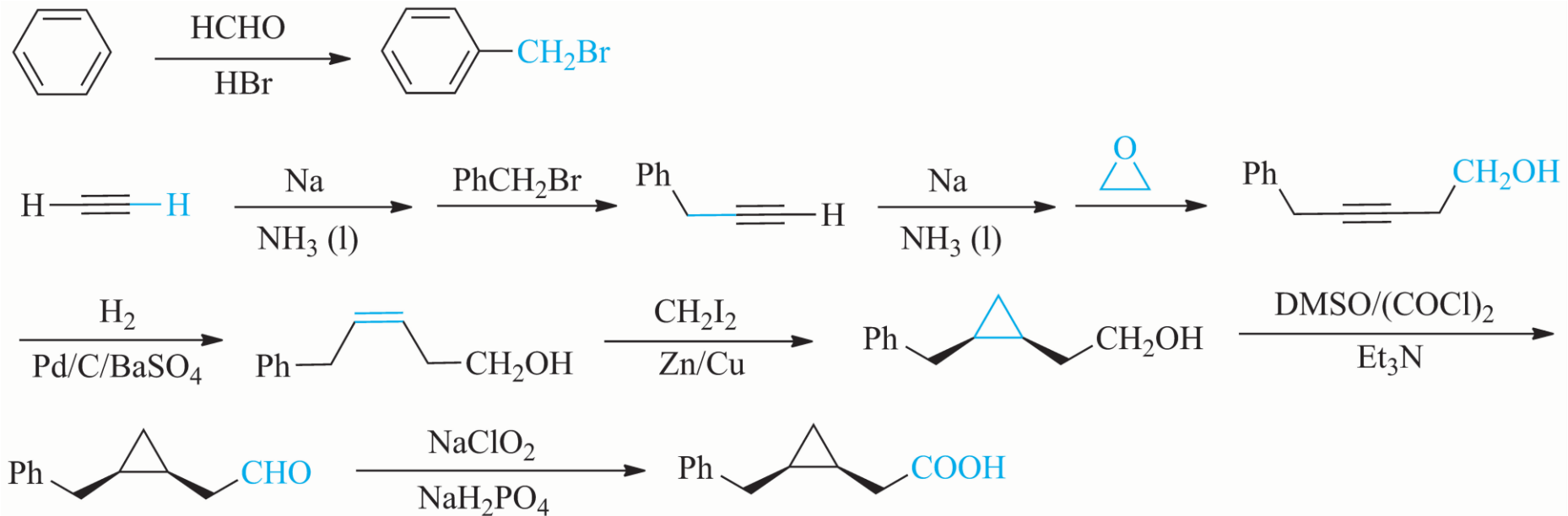


Retrosynthetic analysis:



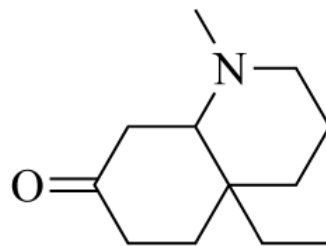


Synthesis:

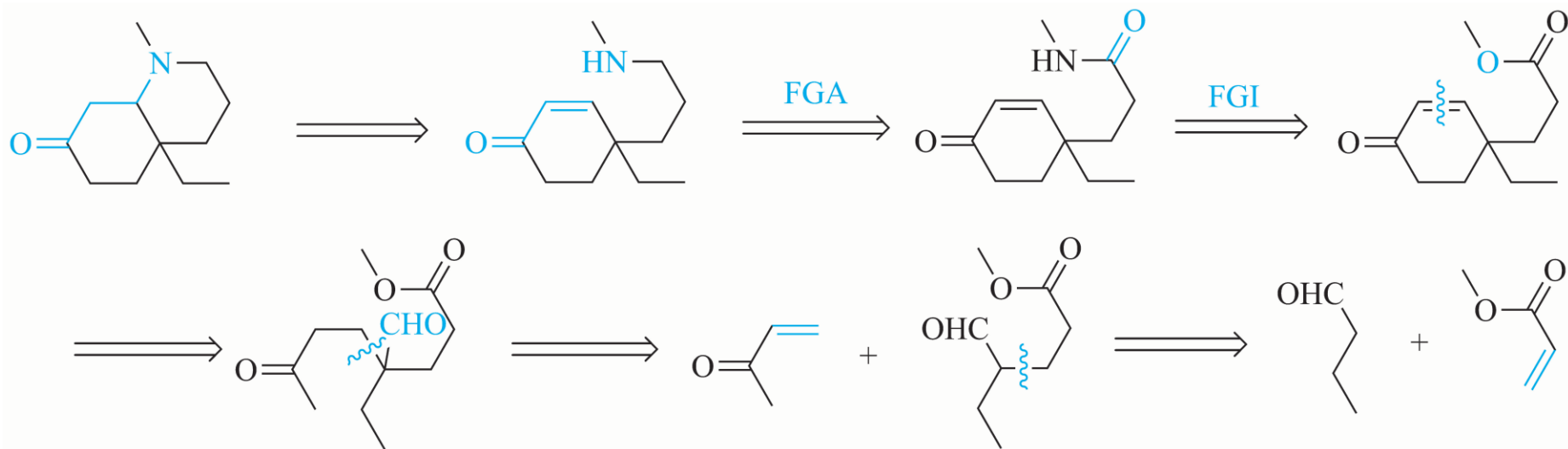




四、以正丁醛和少于四个碳的有机试剂以及必要的无机试剂为原料合成：

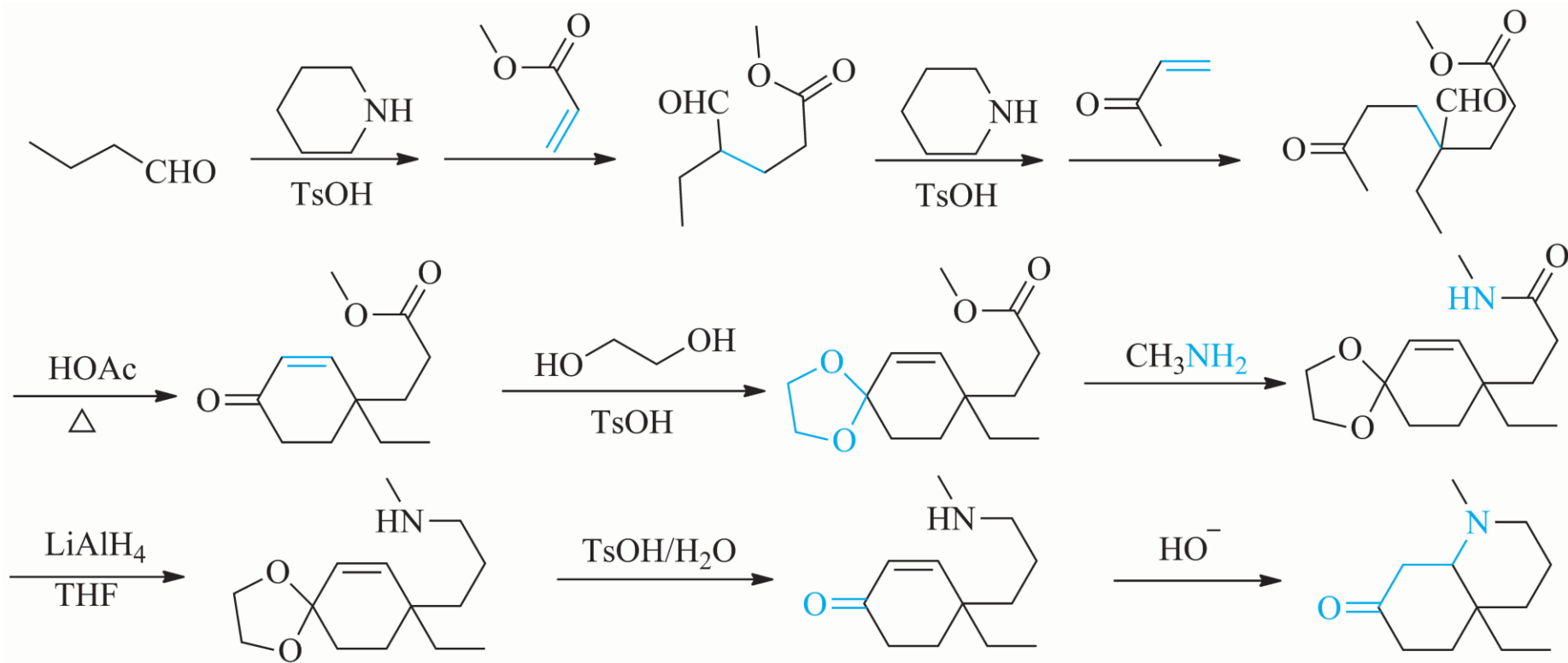


Retrosynthetic analysis:





Synthesis:



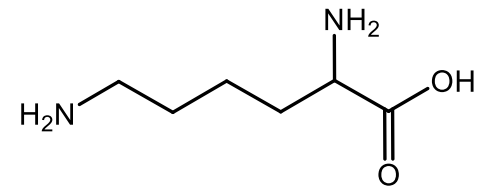
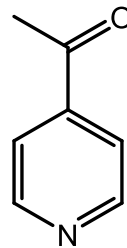
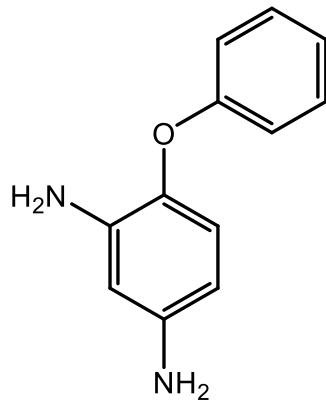
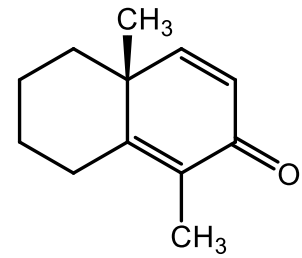
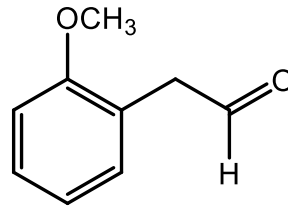
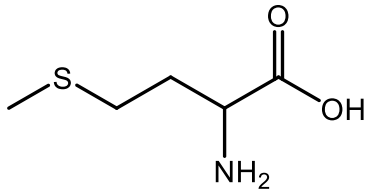


合成设计几个要点

- ▶ **关键的断键**：一正一负，原料易得
举例：1,2-加成；1,4-加成； S_N2 ；环加成；重排反应等
- ▶ **电性选择**（亲电、亲核），电性翻转（拉电子基团、供电子基团）
- ▶ **潜在官能团**（临时存在：保护基、定位基团、占位基团、活性提升基团）
- ▶ **官能团的兼容性**（亲电和亲核，酸性和碱性，氧化性和还原性）



合成练习题





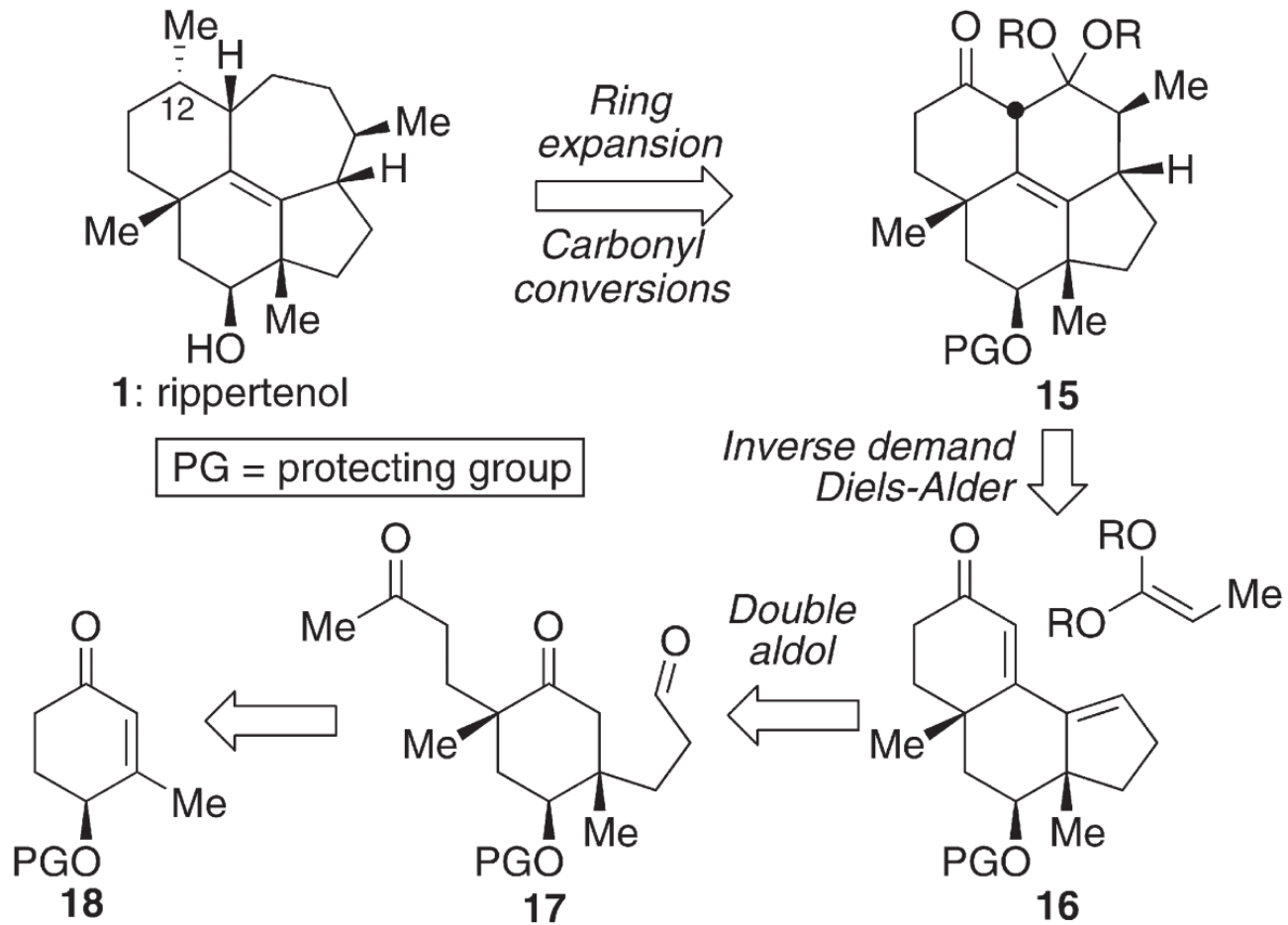
A Concise, Stereocontrolled Total Synthesis of Rippertenol

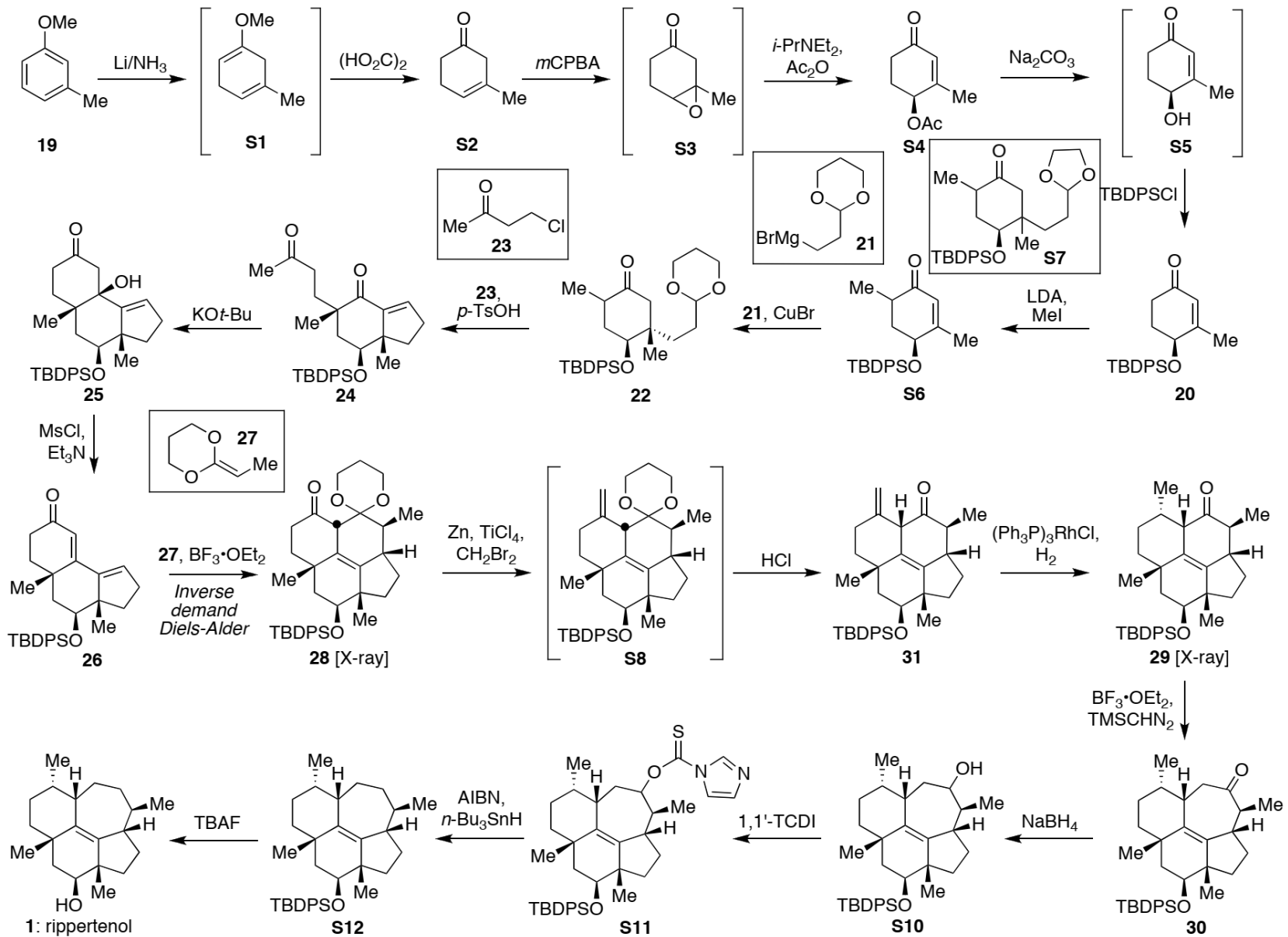
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B: Retrosynthetic analysis







谢谢各位同学!